

BaseRate - Complete Code

Submission source document. Includes the app, trained model, training data, reproducible training script, and verification checks. Line wrapping and Unicode escapes are for print only; use the source package to run the project.

Educational prototype; extensive AI assistance disclosed. No credentials, patient data, or external API keys are included.

dist/index.html

```
1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="utf-8"><meta name="viewport" content="width=device-width,initial-scale=1">
5 <title>BaseRate \u2014 A field guide to screening uncertainty</title>
6 <meta name="description" content="Learn screening uncertainty by counting people, comparing fictional
7 tests and explaining your reasoning to a private on-device AI coach.">
8 <meta name="theme-color" content="#f7f5ef">
9 <link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 32
10 32'%3E%3Crect width='32' height='32' fill='%23f7f5ef'/%3E%3Ctext x='6' y='25' font-family='Georgia'
11 font-size='28' fill='%23242c29'%3E%3C/text%3E%3C/svg%3E">
12 <link rel="stylesheet" href="style.css">
13 <script src="science.js" defer></script><script src="model.js" defer></script><script src="coach.js"
14 defer></script><script src="learning.js" defer></script><script src="app.js" defer></script><script
15 src="lab-tools.js" defer></script><script src="learning-ui.js" defer></script>
16 </head>
17 <body>
18 <a class="skip" href="#lab">Skip to the experiment</a>
19 <header class="site-header"><a class="brand" href="#lab">BaseRate<span>A field guide to screening
20 uncertainty</span></a><nav aria-label="Primary"><a href="#lab">The lab</a><a
21 href="#practice">Practice</a><a href="#coach">AI coach</a><a href="#evidence">Methods</a></nav><a
22 class="header-source" href="https://github.com/Nicholas-afk/baserate" target="_blank"
23 rel="noopener">Source \u2197</a></header>
24 <main>
25 <section class="opening" aria-labelledby="lab-heading"><div><p class="eyebrow">A LEARNING EXPERIMENT /
26 UNIVABIO 2026</p><h1 id="lab-heading">A positive result needs<br><em>a denominator.</em></h1><p
27 class="intro">A test can detect most cases and still produce more false alarms than true positives.
28 Count the people. Change one assumption. See why.</p></div><aside class="opening-note"><span
29 class="small-label">START HERE</span><p>Explore a fictional population below, or predict the answer
30 before you see it.</p><a class="primary-link" href="#practice">Try the three-case lesson <span
31 aria-hidden="true">\u2197</span></a><p class="micro">Fictional rates \xb7 No health details \xb7 No
32 account</p></aside></section>
33 <section id="lab" class="lab" aria-label="Fictional screening experiment">
34 <div class="section-bar"><h2><span class="section-index">01</span> The experiment</h2><div
35 class="bar-actions"><button id="share-experiment" class="text-button">Copy settings
36 link</button><button id="print" class="text-button">Print worksheet</button></div></div>
37 <p class="scope-note">For learning only. These invented rates do not estimate anyone\u2019s health
38 risk or recommend a screening test.</p><p id="link-message" role="status" class="micro"></p>
39 <div class="workspace">
40 <aside class="controls" aria-label="Fictional test settings"><p class="small-label">THREE
41 ASSUMPTIONS</p><label class="scenario-label" for="scenario">Start with</label><select
42 id="scenario"><option value="rare">A rare condition</option><option value="common">A more common
43 condition</option><option value="specific">Fewer false positives</option><option value="custom">Custom
44 experiment</option></select><p id="scenario-note" class="muted">A fictional condition affects 1 in
45 100 people.</p>
46 <div class="control"><label for="prevalence">People with the condition <output for="prevalence"
47 id="prevalence-label">1%</output></label><div class="range-row"><input id="prevalence" type="range"
48 min="0.1" max="50" step="0.1" value="1"><input id="prevalence-number" type="number"
49 aria-label="Prevalence percentage" min="0.1" max="50" step="0.1" value="1"></div><p>Prevalence \xb7
50 the starting group with the condition.</p></div>
51 <div class="control"><label for="sensitivity">Cases the test detects <output for="sensitivity"
52 id="sensitivity-label">90%</output></label><div class="range-row"><input id="sensitivity" type="range"
53 min="1" max="100" step="1" value="90"><input id="sensitivity-number" type="number"
54 aria-label="Sensitivity percentage" min="1" max="100" step="1" value="90"></div><p>Sensitivity \xb7
55 positives among people with the condition.</p></div>
56 <div class="control"><label for="specificity">Non-cases it clears <output for="specificity"
57 id="specificity-label">95%</output></label><div class="range-row"><input id="specificity" type="range"
58 min="1" max="100" step="0.1" value="95"><input id="specificity-number" type="number"
59 aria-label="Specificity percentage" min="1" max="100" step="0.1" value="95"></div><p>Specificity \xb7
60 negatives among people without it.</p></div>
61 <p id="settings-error" role="status" class="error"></p><button class="secondary" id="reset">Reset to
62 the rare case</button><p class="margin-note">Hold the test fixed.<br>Try changing only prevalence.</p>
63 </aside>
64 <div class="population-panel"><div class="panel-top"><div><h3>One thousand people.</h3><p>Four
65 possible outcomes.</p></div><div class="segmented" role="group" aria-label="Population view"><button
66 id="all-view" class="active" aria-pressed="true">All</button><button id="positive-view"
67 aria-pressed="false">Positive</button><button id="tree-view"
68 aria-pressed="false">Tree</button></div></div>
69 <div id="population-wrap"><div id="population" class="population" role="img" aria-label="Population of
70 1000 people"></div><div class="legend"><span><i class="tp"></i>True positive</span><span><i
71 class="fp"></i>False positive</span><span><i class="fn"></i>False negative</span><span><i
72 class="tn"></i>True negative</span></div></div>
73 <div id="frequency-tree" class="frequency-tree" hidden aria-label="Natural frequency tree"></div>
74 <div class="counts"><div><span class="count-tag tp-text">Condition + positive</span><strong
75 id="tp-count">9</strong><small>True positives</small></div><div><span class="count-tag fp-text">No
76 condition + positive</span><strong id="fp-count">49.5</strong><small>False
77 positives</small></div><div><span class="count-tag fn-text">Condition + negative</span><strong
78 id="fn-count">1</strong><small>False negatives</small></div><div><span class="count-tag">No condition
79 + negative</span><strong id="tn-count">940.5</strong><small>True negatives</small></div><p
80 class="rounding-note">Expected counts may be fractional. Squares are rounded; calculations use the
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exact counts.</p>
31 </div>
32 <aside class="result-panel" aria-labelledby="result-heading"><p class="small-label">CHOOSE THE RIGHT
GROUP</p><h3 id="result-heading">Among the people<br>who test positive\u02026</h3><div
class="result-number"><span id="ppv">15.4</span><span>%</span></div><p id="result-explanation"
aria-live="polite">About 9 of 58.5 expected positive results are true positives.</p><div
class="ratio-bar" aria-hidden="true"><span id="ratio-fill"></span></div><div
class="result-key"><span>True</span><span>False</span></div><div class="formula"><span>True positives
\xf7 all positives</span><code id="formula">9 \xf7 (9 + 49.5)</code><small>Multiply by 100 for the
percentage.</small></div><p class="insight" id="insight">A small error rate applied to a large group
can outweigh a high detection rate.</p><a class="text-link" href="#coach">Explain this result
\u2197</a></aside>
33 </div></section>
34 <section id="compare" class="compare-section" aria-labelledby="compare-heading"><div
class="section-bar"><h2 id="compare-heading"><span class="section-index">02</span> Change the starting
population</h2><span class="micro">A counterfactual, not a clinical comparison</span></div><div
class="compare-layout"><div><p class="section-intro">Pin one experiment, then change the lab above.
The test can stay exactly the same while the meaning of a positive result changes.</p><div
class="form-actions"><button id="pin-experiment" class="secondary">Pin current settings as
A</button><button id="compare-common" class="secondary">Compare 1% with 20%</button></div><p
id="pin-status" role="status" class="micro">A is pinned to the rare-condition example. B follows the
lab.</p><div class="table-scroll"><table class="comparison-table"><caption>Expected outcomes in 1,000
people</caption><thead><tr><th scope="col">Measure</th><th scope="col">A \xb7 pinned</th><th
scope="col">B \xb7 current</th></tr></thead><tbody id="compare-rows"><tbody></tbody></table></div><p
id="compare-insight" class="comparison-insight" aria-live="polite"></p></div><div
class="curve-panel"><div class="panel-top"><div><h3>Prevalence changes the answer.</h3><p
id="curve-test-label">The same test, across different populations.</p></div></div><svg
id="prevalence-curve" viewBox="0 0 640 320" role="img" aria-labelledby="curve-title
curve-description"><title id="curve-title">Positive predictive value as prevalence
changes</title><desc id="curve-description">A curve and a current-experiment marker. Equivalent values
are in the table below.</desc></svg><label for="curve-prevalence">Move the current experiment <output
id="curve-value">1%</output></label><input id="curve-prevalence" type="range" min="0.1" max="50"
step="0.1" value="1"><details><summary>Read the curve as a table</summary><table
class="small-table"><thead><tr><th scope="col">Prevalence</th><th scope="col">True among
positive</th></tr></thead><tbody id="curve-table"></tbody></table></details><p class="micro">The line
uses B\u2019s test settings. It starts at 0% prevalence; an undefined value is left
blank.</p></div></div>
35 <details id="sequence" class="sequence-details"><summary>Go further: does a second positive test
settle it?</summary><div class="sequence-body"><div><p class="small-label">THE ASSUMPTION
MATTERS</p><h3>Two positives are not automatically independent evidence.</h3><p>Only people positive
on the first test go through this second step. These rates describe that selected group, not the whole
population.</p><label for="sequence-mode">Choose an assumption</label><select
id="sequence-mode"><option value="independent">Independent, identical test</option><option
value="repeat">Perfectly repeated result</option><option value="custom">Custom conditional
rates</option></select><p id="sequence-assumption" class="assumption-note">Independence is an invented
assumption here, not evidence about a real test.</p><div class="sequence-controls"><div><label
for="conditional-detection">Second-positive among true first-positives <output
id="conditional-detection-label">90%</output></label><input id="conditional-detection" type="range"
min="0" max="100" step="1" value="90"></div><div><label for="conditional-false">Second-positive among
false first-positives <output id="conditional-false-label">5%</output></label><input
id="conditional-false" type="range" min="0" max="100" step="0.1" value="5"></div></div><div
class="sequence-result"><p class="small-label">AMONG TWO POSITIVE RESULTS</p><strong
id="sequence-ppv">76.6%</strong><p id="sequence-counts"></p><div id="sequence-steps"
class="sequence-steps"></div><p class="micro">A second step can lose true positives as well as remove
false positives. This is a mathematical demonstration, not a recommendation to repeat a real
test.</p></div></div></details></section>
36 <section id="practice" class="practice-section" aria-labelledby="practice-heading"><div
class="section-bar"><h2 id="practice-heading"><span class="section-index">03</span> Predict. Count.
Reconsider.</h2><span id="practice-progress" class="micro">Case 1 of 3</span></div><p
class="section-intro">Three short experiments. Make an estimate before revealing the arithmetic, then
name the group in the denominator. This is practice, not a test of your health knowledge.</p><div
class="practice-layout"><ol id="practice-steps" class="practice-steps" aria-label="Lesson
progress"></ol><div><div id="practice-case"><p id="case-settings" class="small-label"></p><h3
id="case-title" tabindex="-1"></h3><p id="case-prompt"></p><form id="prediction-form"
novalidate><label for="prediction">Your estimate: true positives among positive results</label><div
class="prediction-input"><input id="prediction" type="number" min="0" max="100" step="any" required
placeholder="0\u2013100" inputmode="decimal"><span>%</span></div><button id="predict-button"
class="primary">Reveal the counts</button></div><p id="prediction-error" role="status"
class="error"></p></form><div id="case-reveal" class="case-reveal" tabindex="-1" hidden><h4
id="prediction-result"></h4><p id="prediction-difference"></p><code id="case-equation"
class="case-equation"></code><p id="case-explanation"></p></div><form
id="denominator-form"><fieldset><legend>Which group goes in the denominator?</legend><label><input
type="radio" name="denominator" value="cases"> Everyone with the condition</label><label><input
type="radio" name="denominator" value="positives"> Everyone who tests positive</label><label><input
type="radio" name="denominator" value="population"> All 1,000 people</label></fieldset></form><p
id="denominator-feedback" role="status"></p><button id="practice-next" class="primary" disabled>Next
case</button></div></div><div id="practice-complete" tabindex="-1" hidden><p class="small-label">YOUR
LEARNING NOTEBOOK</p><h3>What changed in your estimates?</h3><p>Your first guesses are kept for this
tab only. A smaller gap on these cases is not proof of learning or a clinical result. Try explaining
the change in your own words to the coach.</p><div class="table-scroll"><table
class="small-table"><thead><tr><th scope="col">Case</th><th scope="col">Guess</th><th
scope="col">Actual</th><th scope="col">Gap</th><th scope="col">Denominator</th></tr></thead><tbody
id="notebook-rows"></tbody></table></div><div class="form-actions"><button id="download-notebook"
class="secondary">Export my notebook</button><button id="restart-practice" class="secondary">Start
again</button></div></div></section>
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again</button><a href="#coach">Explain the lesson \u2197</a></div></div></div></div><p
class="micro">Nothing is saved or sent. An export includes numeric guesses and selected options, never
your coach explanation.</p></section>
37 <section id="coach" class="coach-section" aria-labelledby="coach-heading"><div class="section-bar"><h2
id="coach-heading"><span class="section-index">04</span> Put the reasoning into words</h2><span
class="micro">AI runs on this device</span></div><p class="section-intro">Explain the current lab
result. A small trained classifier looks for a reasoning pattern and selects an evidence-linked
response. It may misread your wording.</p><div class="coach-layout"><form id="coach-form"><label
for="explanation">Your explanation of this experiment</label><textarea id="explanation"
maxlength="1800" rows="5" aria-describedby="coach-help" placeholder="A positive result doesn\u2019t
prove the condition because\u2026"></textarea><p class="muted" id="coach-help">Fictional examples
only. English in this prototype. Your words stay in this browser and disappear on reload.</p><div
class="form-actions"><button type="submit" class="primary">Check my reasoning</button><button
id="example" type="button" class="secondary">Use an example</button></div></form><div
id="coach-response" class="coach-response" aria-live="polite"><p class="small-label">A SECOND LOOK AT
YOUR REASONING</p><h3>Start with the positive group.</h3><p>How can a rare condition and a small
false-positive rate combine to produce this result?</p><p class="muted">Model feedback is a
suggestion. The arithmetic is calculated separately and does not depend on
AI.</p></div></div></section>
38 <section id="evidence" class="evidence-section" aria-labelledby="evidence-heading"><div
class="section-bar"><h2 id="evidence-heading"><span class="section-index">05</span> Methods, evidence
& honest limits</h2><a href="project.html">The project file \u2197</a></div><div
class="evidence-grid"><article><p class="small-label">REPRESENTATION</p><h3>Show the people in the
percentage.</h3><p>Natural frequencies make the relevant groups visible. The square grid and frequency
tree are two views of the same expected counts. The result uses Bayes\u2019 theorem; population size
changes counts, not proportions.</p><a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC4604268/"
target="_blank" rel="noopener">Hoffrage et al., 2015 \u2197 original study \u2197</a><p
class="micro">Research informed the representation. BaseRate itself has not had a learner
study.</p></article><article><p class="small-label">SCREENING</p><h3>Every test has a
context.</h3><p>False positives and missed cases matter. These three rates cannot establish the
benefits, harms or suitability of a real screening program. There is no personal diagnosis, risk
estimate or treatment advice here.</p><a
href="https://www.cancer.gov/about-cancer/screening/patient-screening-overview-pdq" target="_blank"
rel="noopener">National Cancer Institute \u2197 overview \u2197</a></article><article><p
class="small-label">MODEL CARD</p><h3>A small model you can inspect.</h3><p
id="model-summary"></p><div class="model-details"><div class="model-summary">Read training, evaluation and failure modes</div><div
id="model-details"><p class="micro">Extensive AI assistance was used for code,
synthetic examples and educational copy. Your explanation is never sent to a generative
model.</p></div><div class="methods-details"><div class="methods-content"><p>For population N, prevalence p, sensitivity s and
specificity c, expressed as proportions:</p><code>TP = N \u2197 p \u2197 s<br>FP = N \u2197 (1 \u2197 p)
\u2197 (1 \u2197 c)<br>FN = N \u2197 p \u2197 (1 \u2197 s)<br>TN = N \u2197 (1 \u2197 p) \u2197 c<br>PPV = TP
\u2197 (TP + FP)</code><p>For a second test among first-positives: both-positive true cases = TP \u2197
conditional detection; both-positive non-cases = FP \u2197 conditional false-positive rate. Reusing
unconditional rates requires a conditional-independence assumption. A zero denominator has no defined
proportion.</p><p>Only visual squares are rounded. Expected counts, comparisons, curves and practice
answers retain the unrounded values.</p></div></div></div></div>
39 </main><footer><a class="brand" href="#lab">BaseRate</a><p>UnivaBio 2026 \u2197 Educational
prototype<br>Fictional populations. Transparent assumptions.</p><div><a href="project.html">Project
materials</a><a href="https://github.com/Nicholas-afk/baserate" target="_blank" rel="noopener">Code &
datasets \u2197</a></div></div>
40 </body></html>

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dist/project.html

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1 <!doctype html><html lang="en"><head><meta charset="utf-8"><meta name="viewport"
content="width=device-width,initial-scale=1"><title>BaseRate \xb7 The project file</title><link
rel="stylesheet" href="style.css"><style>.materials{max-width:980px;margin:50px auto;padding:0
24px}.materials h1{font-size:clamp(36px,6vw,64px);margin:12px 0 25px}.materials h2{font:28px
Georgia;margin:35px 0 16px}.materials p{line-height:1.65}.materials
video{display:block;width:100%;margin:24px
0;background:#242c29}.material-links{display:flex;flex-wrap:wrap;gap:12px;margin:24px
0}.material-links a{padding:10px 14px;border:1px solid
#81877c;font-size:13px;text-decoration:none}.materials li{margin:12px
0;font-size:14px}</style></head><body><header class="site-header"><a class="brand"
href="index.html">BaseRate</a><nav aria-label="Project navigation"><a
href="index.html">Open the learning lab \u2197</a><a
href="https://github.com/Nicholas-afk/baserate">Source \u2197</a></nav></header><main
class="materials"><p class="eyebrow">UNIVABIO 2026 / AI FOR HUMAN HEALTH</p><h1>Understand the
denominator.</h1><p>An interactive field guide to screening uncertainty. Explore fictional
populations, compare experiments, inspect the assumptions behind repeat testing, predict before
revealing the counts, and explain the result to an on-device AI coach.</p><div
class="material-links"><a href="index.html">Try the lab</a><a
href="downloads/BaseRate-Project-Description.pdf">One-page description</a><a
href="downloads/BaseRate-Complete-Code.pdf">Complete code PDF</a><a
href="downloads/BaseRate-Source.zip">Runnable source ZIP</a><a
href="https://github.com/Nicholas-afk/baserate">Repository, data & tests \u2197</a></div><h2>Working
prototype demonstration</h2><video controls preload="metadata" aria-label="BaseRate demonstration
narrated by the entrant"><source src="downloads/BaseRate-Demo.mp4"
type="video/mp4"></video><p>Captured states of the actual application, narrated using the
entrant\u2019s supplied voice recording. The interface and educational copy were created with
extensive AI assistance.</p><h2>What you can investigate</h2><ul><li>Keep detection and clearance
fixed while changing prevalence. Pin experiment A, change B, and compare their positive groups and a
continuous prevalence curve.</li><li>Switch from population squares to a natural-frequency tree. The
table and tree retain raw expected counts; only squares are rounded.</li><li>Explore a second test
among first-positive people. Choose independent tests, perfectly repeated results, or explicit
conditional rates. A repeated result can add no information.</li><li>Complete three
predict/reveal/denominator cases and export a numeric learning notebook. First guesses stay in memory
for this tab only; learner explanations are never exported.</li><li>Inspect influential model
features, label alternatives and uncalibrated votes. These are model diagnostics, not health
probabilities.</li></ul><h2>Verification and limitations</h2><ul><li>336 arithmetic and
population-rounding combinations; 2,525 prevalence-curve points; 405 conditional repeat-test
combinations; separate practice and export checks.</li><li>100 synthetic English training
explanations. On a separate 30-example synthetic holdout: 29 correct raw classifications; uncertainty
checks accepted 28 labels, of which 27 were correct.</li><li>Known error: one correct distinction
between sensitivity and predictive value was misread. Negation, mixed ideas, unfamiliar wording and
other languages can fail. Votes are uncalibrated.</li><li>No independent learner study, clinical
validation or measured health benefit. All scenarios are fictional; no personal diagnosis, risk
estimate or treatment advice.</li></ul><h2>Privacy and authorship</h2><p>No patient data, analytics,
remote inference, external scripts or fonts. The app stores no learner text and makes no inference
requests. Hosting providers may retain ordinary request logs. The code, synthetic dataset, educational
copy and submission materials were created with extensive AI assistance at the entrant\u2019s
direction. The entrant supplied the narration recording and must understand and explain the work to
judges.</p><h2>Reproduce the work</h2><p>The repository includes the complete static app, synthetic
data, standard-library Python trainer, evaluation details and Node verification checks. The source ZIP
runs locally without app dependencies. The one-page description states what was built, what was
verified and what remains unvalidated.</p></main><footer><a class="brand"
href="index.html">BaseRate</a><p>Educational prototype \xb7 Fictional screening
scenarios</p></footer></body></html>
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dist/style.css

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1 :root{--paper:#f7f5ef;--white:#fffefa;--ink:#242c29;--muted:#626760;--line:#d7d7ca;--strong:#81877c;--green:#245b4a;--rust:#944a2d;--amber:#b67925;--rose:#9e5065;--wash:#eeeeee4;--sans:ui-sans-serif,-apple-system,BlinkMacSystemFont,"Segoe UI",sans-serif;--serif:Georgia,"Times New Roman",serif}*{box-sizing:border-box}html{scroll-behavior:smooth;scroll-padding-top:24px}body{margin:0;background:var(--paper);color:var(--ink);font:16px/1.6 var(--sans)}button,input,select,textarea{font:inherit;color:var(--green);text-decoration:underline:4px;text-decoration-thickness:1px;a:hover{text-decoration-thickness:2px}button,a,input,select,textarea{touch-action:manipulation}button{cursor:pointer}button:disabled{cursor:default;opacity:.5}button:focus-visible,a:focus-visible,input:focus-visible,textarea:focus-visible,select:focus-visible,summary:focus-visible,[tabindex="-1"]:focus-visible{outline:3px solid var(--rust);outline-offset:4px}[hidden]{display:none!important}.skip{position:absolute;top:-100px;left:20px;background:var(--white);z-index:10;padding:12px}.skip:focus{top:10px}.site-header{display:flex;align-items:center;gap:36px;max-width:1560px;margin:auto;padding:24px 4%;border-bottom:1px solid var(--line)}.brand{font:700 30px/1.1 var(--serif);letter-spacing:-1.2px;color:var(--ink);text-decoration:none;white-space:nowrap}.brand span{display:block;font:11px/1.6 var(--sans);letter-spacing:.2px;color:var(--muted);margin-top:7px}nav{display:flex;gap:27px;margin-left:auto}nav a,.header-source{font-size:14px;text-decoration:none;color:var(--ink)}nav a:hover,.header-source:hover{text-decoration:underline}.header-source{border-left:1px solid var(--line);padding-left:27px}main{max-width:1560px;margin:auto;padding:0 4%}.opening{display:grid;grid-template-columns:1fr 300px;gap:50px;align-items:center;padding:45px 0 30px}.eyebrow,.small-label{font:600 11px/1.5 var(--sans);letter-spacing:1.2px;margin:0 0 14px;color:var(--muted)}h1{font:400 4.2vw,64px}/1.05 var(--serif);letter-spacing:-1.8px;margin:0}h1 em{font-weight:400;color:var(--green)}.intro{max-width:680px;color:var(--muted);margin:19px 0 0;font-size:16px;line-height:1.65}.opening-note{border-left:1px solid var(--line);padding-left:26px}.opening-note p{font-size:14px;margin:0 0 16px}.primary-link{display:flex;justify-content:space-between;border-bottom:1px solid var(--green);padding-bottom:8px;text-decoration:none;font-size:14px;font-weight:600}.opening-note .micro{font-size:11px;margin-top:14px}.section-bar{display:flex;justify-content:space-between;align-items:center;gap:24px;border-top:2px solid var(--ink);padding:17px 0 0}.section-bar h2{font:400 25px/1.3 var(--serif);letter-spacing:-.4px;margin:0}.section-index{font:500 13px var(--sans);color:var(--muted);padding-right:12px}.bar-actions{display:flex;gap:20px}.text-button{border:0;background:none;color:var(--green);font-size:13px;padding:5px 0;text-decoration:underline;text-decoration:underline:4px}.scope-note{font-size:12px;color:var(--muted);margin:8px 0 18px}.micro,.rounding-note{font-size:12px;color:var(--muted);line-height:1.6}.micro.empty,.error.empty{display:none}.workspace{display:grid;grid-template-columns:245px minmax(300px,1fr) 260px;border:1px solid var(--line);background:var(--white)}.controls{padding:25px 22px;border-right:1px solid var(--line)}.scenario-label{display:block;font-size:13px;font-weight:600;margin-bottom:7px}select{width:100%;padding:10px 9px;border:1px solid var(--strong);border-radius:2px;background:var(--white);color:var(--ink);font-size:14px}.muted{font-size:13px;color:var(--muted)}.control{margin:25px 0}.control label{display:flex;justify-content:space-between;align-items:center;gap:8px;font-size:13px;line-height:1.5;font-weight:600}.control output{color:var(--green);font-size:17px;font-variant-numeric:tabular-nums}.range-row{display:flex;align-items:center;gap:10px;margin-top:9px}input[type=range]{accent-color:var(--green);min-height:30px;width:100%;margin:0}input[type=number]{font-variant-numeric:tabular-nums;border:1px solid var(--strong);background:var(--white);color:var(--ink);padding:7px 6px;border-radius:2px;min-height:38px}.range-row input[type=number]{width:65px;flex-shrink:0;font-size:13px}.control p{font-size:12px;color:var(--muted);line-height:1.5;margin:6px 0 0}.secondary,.primary{padding:10px 15px;min-height:42px;border-radius:2px;font-size:13px;font-weight:600}.secondary{border:1px solid var(--strong);background:transparent;color:var(--ink)}.secondary:hover{background:var(--wash)}.primary{border:1px solid var(--green);background:var(--green);color:white}.primary:hover{background:#173e32}.margin-note{border-top:1px solid var(--line);padding-top:16px;margin:22px 0 0;font:italic 17px/1.5 var(--serif);color:var(--muted)}.error{font-size:13px;color:#9c3529}.population-panel{padding:25px 24px;min-width:0}.panel-top{display:flex;justify-content:space-between;align-items:flex-start;gap:18px;margin-bottom:20px}h3{font:400 24px/1.3 var(--serif);letter-spacing:-.4px;margin:0}.panel-top h3{font-size:22px}.panel-top p{font-size:12px;color:var(--muted);margin:5px 0 0}.segmented{display:flex;align-items:center;border:1px solid var(--strong);border-radius:2px}.segmented button{font-size:12px;padding:8px 10px;background:transparent;border:0;border-right:1px solid var(--line);color:var(--ink)}.segmented button:last-child{border-right:0}.segmented button.active{background:var(--ink);color:var(--white)}.population{display:grid;grid-template-columns:repeat(40,1fr);gap:3px;align-content:start;min-height:280px}.person{width:100%;aspect-ratio:1;border-radius:0}.person.tp,.legend .tp{background:var(--green)}.person.fp,.legend .fp{background:repeating-linear-gradient(135deg,var(--amber) 0 3px,#d8b47f 3px 4px)}.person.fn,.legend .fn{background:var(--rose);box-shadow:inset 0 0 2px #d5a7b2}.person.tn,.legend .tn{background:#e3e3d9;border:1px solid #c6cabc}.person.hidden{opacity:.12}.legend{display:flex;flex-wrap:wrap;gap:8px 14px;color:var(--muted);font-size:11px;margin-top:16px}.legend span{display:flex;align-items:center;gap:6px}.legend i{width:10px;height:10px}.counts{display:grid;grid-template-columns:repeat(4,1fr);gap:10px;border-top:1px solid var(--line);padding-top:17px;margin-top:22px}.counts>div{display:flex;flex-direction:column}.count-tag{font-size:10px;font-weight:500;line-height:1.4}.counts strong{font-size:27px;font-weight:500;letter-spacing:-1px;font-variant-numeric:tabular-nums;margin-top:4px}.counts small{font-size:11px;color:var(--muted)}.tp-text{color:var(--green)}.fp-text{color:#8b5417}.fn-text{color:#944359}.rounding-note{font-size:11px;margin:18px 0 0}.result-panel{border-left:1px solid var(--line);padding:25px 23px;background:#eeeeee4}.result-panel h3{font-size:24px;line-height:1.3}.result-number{display:flex;align-items:baseline;margin:23px 0 6px;gap:2px;color:var(--green);font-variant-numeric:tabular-nums}.result-number span:first-child{font:400 70px/1 var(--serif);letter-spacing:-3px}.result-number span:last-child{font:400 29px var(--serif)}.result-panel>p:not(.small-label){font-size:13px}.ratio-bar{height:11px;background:var(--amber);margin-top:18px;overflow:hidden}.ratio-bar span{display:block;hei
```

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ght:100%;background:var(--green)}.result-key{display:flex;justify-content:space-between;font-size:10px
;color:var(--muted);margin-top:4px}.formula{border-top:1px solid var(--line);border-bottom:1px solid
var(--line);padding:16px 0;margin:20px 0}.formula span,.formula
small{display:block;font-size:11px;color:var(--muted)}.formula
code{display:block;font-size:17px;margin:4px 0;font-variant-numeric:tabular-nums}.text-link{display:bl
ock;font-size:13px;font-weight:600;margin-top:23px}.frequency-tree{min-height:310px}.tree-root{border-
bottom:1px solid var(--ink);font:21px var(--serif);padding:5px 0
16px;margin-bottom:20px}.tree-branches{display:grid;grid-template-columns:1fr
1fr;gap:22px}.tree-branch{border-left:1px solid
var(--strong);padding-left:16px}.tree-parent{font-size:12px;margin-bottom:20px}.tree-parent strong{fon
t-size:23px;display:block;font-weight:500}.tree-leaf{display:flex;justify-content:space-between;align-
items:center;gap:10px;font-size:12px;padding:12px 0;border-top:1px solid var(--line)}.tree-leaf b{font
-size:18px;font-weight:500}.compare-section,.practice-section,.coach-section,.evidence-section{padding
:48px 0 0}.section-intro{font-size:15px;color:var(--muted);max-width:780px;margin:18px 0
22px}.compare-layout{display:grid;grid-template-columns:1fr
1fr;gap:48px;margin-top:20px}.compare-layout .section-intro{margin-top:0}.form-actions{display:flex;al
ign-items:center;gap:10px;flex-wrap:wrap;margin-top:16px}.table-scroll{overflow-x:auto}table{width:100
%;border-collapse:collapse;font-variant-numeric:tabular-nums;font-size:13px;text-align:left}caption{te
xt-align:left;color:var(--muted);font-size:12px;margin-bottom:8px}th{font-weight:600}th,td{border-bott
om:1px solid var(--line);padding:10px
8px}th:first-child,td:first-child{padding-left:0}thead{border-top:1px solid
var(--strong)}.comparison-table td:not(:first-child){font-weight:500}.comparison-table tr:last-child
td{font-size:23px;color:var(--green);font-family:var(--serif)}.comparison-insight{border-left:2px
solid var(--green);padding-left:14px;font-size:14px}.curve-panel{border-left:1px solid
var(--line);padding-left:35px}.curve-panel svg{width:100%;height:auto;display:block}.curve-panel label
{display:flex;justify-content:space-between;font-size:13px;font-weight:600;margin-bottom:6px}.curve-pa
nel output{color:var(--green)}.curve-grid{stroke:var(--line);stroke-width:1}.curve-axis-label{fill:var
(--muted);font:13px var(--sans)}.curve-line{stroke:var(--green);stroke-width:3;fill:none}.curve-guide{
stroke:var(--strong);stroke-width:1;stroke-dasharray:3 4}.curve-dot{fill:var(--green);stroke:var(--whi
te);stroke-width:3}.curve-marker-label{fill:var(--green);font:600 15px
var(--sans)}summary{cursor:pointer;color:var(--green);font-size:13px;padding:8px 0;text-underline-offs
et:4px}details{margin-top:13px}details[open]>summary{margin-bottom:12px}.sequence-details{border-top:1
px solid var(--strong);border-bottom:1px solid var(--line);margin-top:28px;padding:7px
0}.sequence-details>summary{font:400 22px var(--serif);color:var(--ink);padding:13px
0}.sequence-body{display:grid;grid-template-columns:1.15fr 1fr;gap:48px;padding:10px 0
25px}.sequence-body p{font-size:14px}.sequence-body
label{font-size:13px;font-weight:600;display:block}.sequence-body
select{max-width:360px;margin-top:7px}.assumption-note{color:var(--rust);border-left:2px solid
var(--rust);padding-left:12px}.sequence-controls{display:grid;grid-template-columns:1fr
1fr;gap:23px}.sequence-controls label{font-size:12px;line-height:1.5}.sequence-controls output{display
:block;color:var(--green);font-size:17px;margin-top:5px}.sequence-result{background:var(--wash);padding:
25px}.sequence-result>strong{font:400 58px
var(--serif);color:var(--green)}.sequence-steps{border-top:1px solid
var(--line);padding-top:12px}.sequence-steps p{margin:6px
0;font-size:13px}.practice-layout{display:grid;grid-template-columns:270px 1fr;gap:46px;padding:22px 0
25px}.practice-steps{list-style:none;margin:0;padding:0;border-top:1px solid
var(--line)}.practice-steps li{border-bottom:1px solid var(--line);padding:18px
0;font-size:13px;color:var(--muted)}.practice-steps
li.current{color:var(--green);font-weight:600}.practice-steps li.done:after{content:'
\u2713';color:var(--green)}#practice-case,#practice-complete{max-width:820px}.practice-layout
h3{font-size:31px}.practice-layout label{font-size:13px}.prediction-input{display:flex;align-items:cen
ter;gap:10px;margin-top:8px}.prediction-input input{width:110px;font-size:20px;padding:8px
12px}.prediction-input .primary{margin-left:10px}.case-reveal{border-top:1px solid
var(--line);padding-top:20px;margin-top:25px}.case-reveal h4{font:400 27px/1.3
var(--serif);color:var(--green);margin:0}.case-reveal
p{font-size:14px}.case-equation{display:block;padding:13px
15px;background:var(--wash);font-size:16px;line-height:1.6;overflow-wrap:anywhere}.case-reveal
fieldset{margin:20px 0 0;padding:0;border:0}.case-reveal
legend{font-size:14px;font-weight:600;margin-bottom:10px}.case-reveal fieldset
label{display:block;margin:9px 0}input[type=radio]{accent-color:var(--green);width:17px;height:17px;ve
rtical-align:middle;margin-right:6px}.coach-layout{display:grid;grid-template-columns:1fr
1fr;gap:46px}.coach-layout
label{display:block;font-size:14px;font-weight:600;margin-bottom:12px}textarea{width:100%;border:1px
solid var(--strong);background:var(--white);min-height:160px;resize:vertical;border-radius:2px;padding
:15px;line-height:1.6;color:var(--ink);font-size:16px}.coach-response{background:var(--wash);border-le
ft:2px solid var(--green);padding:25px 28px}.coach-response h3{font-size:28px}.coach-response
p{font-size:14px}.coach-icon{display:none}.coach-response
.small-label{font-size:11px;color:var(--green)}.coach-response
.muted{font-size:12px}.feedback-source{border-top:1px solid var(--line);padding-top:13px;margin-top:20
px;font-size:12px}.feedback-tags{display:flex;gap:8px;flex-wrap:wrap}.feedback-tags span{padding:3px
7px;border:1px solid var(--strong);font-size:11px}.coach-response details{border-top:1px solid var(--l
ine);padding-top:5px}.diagnostic-bars{margin-top:13px}.diagnostic-row{display:grid;grid-template-colum
ns:105px 1fr 48px;gap:10px;align-items:center;font-size:11px;margin:7px
0}.diagnostic-meter{height:5px;background:#d8dacf}.diagnostic-meter span{display:block;height:100%;bac
kground:var(--green)}.evidence-section{padding-bottom:45px}.evidence-grid{display:grid;grid-template-c
olumns:1fr 1fr 1fr;gap:40px;padding-top:25px}.evidence-grid article{border-right:1px solid
var(--line);padding-right:30px}.evidence-grid
article:last-child{border-right:0;padding-right:0}.evidence-grid h3{font-size:25px}.evidence-grid
p{font-size:14px;color:var(--muted)}.evidence-grid .small-label,.evidence-grid
.micro{font-size:11px}.evidence-grid a{font-size:12px}.model-metric{color:var(--ink)!important;font-we
ight:600}.methods-details{border-top:1px solid var(--line);margin-top:30px;padding-top:8px}.methods-co
ntent{max-width:850px;font-size:14px}.methods-content
code{display:block;background:var(--wash);padding:18px}.methods-content
p{color:var(--muted)}footer{border-top:1px solid var(--line);padding:27px 4% 35px;max-width:1560px;mar

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gin:auto;display:flex;justify-content:space-between;align-items:center;gap:30px}footer
.brand{font-size:27px}footer
p{font-size:12px;color:var(--muted);margin:0}footer>div{display:flex;gap:22px;font-size:12px}
2 @media(min-width:1450px){.workspace{grid-template-columns:270px minmax(400px,1fr)
285px}.population{min-height:350px;gap:4px}.result-panel{padding:27px}.result-number
span:first-child{font-size:80px}}
3 @media(max-width:1150px){.workspace{grid-template-columns:230px
minmax(300px,1fr)}.result-panel{grid-column:1/-1;border-left:0;border-top:1px solid
var(--line);display:grid;grid-template-columns:1fr 1fr 1fr;gap:0
28px}.result-panel>.small-label{grid-column:1}.result-panel h3{grid-column:1;grid-row:2}.result-number
{grid-column:2;grid-row:1/3;margin:0;align-self:center}.result-panel>#result-explanation{grid-column:3
;grid-row:1/3}.result-panel .ratio-bar,.result-panel
.result-key{display:none}.formula{grid-column:1;grid-row:3;margin:15px 0
0;border-bottom:0}.insight{grid-column:2;grid-row:3}.result-panel
.text-link{grid-column:3;grid-row:3;align-self:center}.opening{grid-template-columns:1fr 240px;gap:30p
x}.compare-layout{gap:25px}.curve-panel{padding-left:25px}.header-source{display:none}.practice-layout
{grid-template-columns:220px 1fr;gap:30px}.sequence-body{gap:30px}}
4 @media(max-width:780px){.site-header{gap:20px;flex-wrap:wrap;padding:22px 5% 15px}.site-header
nav{margin-left:0;width:100%;gap:0;justify-content:space-between}.site-header nav
a{font-size:13px}.brand{font-size:29px}.brand span{font-size:11px}main{padding:0
5%}.opening{grid-template-columns:1fr;padding:30px 0;gap:25px}h1{font-size:43px;letter-spacing:-1.3px}
.intro{font-size:15px}.opening-note{border-left:0;border-top:1px solid var(--line);padding:15px 0
0;max-width:460px}.opening-note .small-label{display:none}.opening-note
p{font-size:13px;margin-bottom:12px}.opening-note .micro{font-size:11px;margin-bottom:0}.section-bar{f
lex-wrap:wrap;gap:8px;padding-top:13px}.section-bar h2{font-size:24px}.bar-actions{gap:18px}.scope-not
e{font-size:11px}.workspace{grid-template-columns:1fr}.controls{border-right:0;border-bottom:1px solid
var(--line);padding:22px}.controls>.small-label{margin-bottom:9px}.control{margin:20px 0}.control
label{font-size:14px}.control p{font-size:12px}.range-row
input[type=number]{width:75px;font-size:15px}.margin-note{display:none}.population-panel{padding:22px
17px}.panel-top{flex-wrap:wrap;gap:10px}.population{grid-template-columns:repeat(25,1fr);gap:3px;min-h
eight:420px}.counts{grid-template-columns:repeat(2,1fr);gap:14px}.count-tag{font-size:12px}.counts
small{font-size:12px}.counts strong{font-size:31px}.legend{font-size:11px}.result-panel{grid-column:au
to;display:block;padding:25px;border-top:1px solid var(--line)}.result-panel h3
br{display:none}.result-number{margin:18px 0}.result-panel .ratio-bar,.result-panel
.result-key{display:block}.result-panel .result-key{display:flex}.formula{margin:20px
0;border-bottom:1px solid var(--line)}.result-panel .text-link{margin-top:15px}.frequency-tree{min-hei
ght:300px}.tree-branches{gap:15px}.tree-branch{padding-left:10px}.compare-section,.practice-section,.c
oach-section,.evidence-section{padding-top:35px}.compare-layout,.sequence-body,.coach-layout,.evidence
-grid{grid-template-columns:1fr;gap:25px}.curve-panel{border-left:0;border-top:1px solid
var(--line);padding:23px 0 0}.sequence-controls{gap:20px}.sequence-details>summary{font-size:21px;line
-height:1.4}.sequence-result{padding:22px}.practice-layout{grid-template-columns:1fr;gap:23px;padding-
top:5px}.practice-steps{display:flex;gap:12px;border-top:0}.practice-steps
li{flex:1;font-size:11px;padding:12px 0;border-top:1px solid
var(--line);border-bottom:0}.practice-layout
h3{font-size:28px}.prediction-input{flex-wrap:wrap}.prediction-input
.primary{margin-left:0}.coach-response{padding:22px}.evidence-grid{padding-top:22px}.evidence-grid
article{border-right:0;border-bottom:1px solid var(--line);padding:0 0 20px}.evidence-grid article:las
t-child{border-bottom:0}.section-intro{font-size:14px}.case-equation{font-size:14px}footer{align-items
:flex-start;flex-wrap:wrap;padding:25px
5%}footer>div{width:100%}table{font-size:12px}th,td{padding:9px 6px}}
5 @media(prefers-reduced-motion:reduce){html{scroll-behavior:auto}}
6 @media print{body{background:white;color:black;font-size:11px}.site-header nav,.header-source,.opening
-note,.bar-actions,.compare-section,.practice-section,.coach-section,.evidence-section,footer,button{d
isplay:none}.opening{padding:15px 0}.opening h1{font-size:34px}.workspace{grid-template-columns:180px
1fr 180px}.controls,.population-panel,.result-panel{padding:12px}.result-panel{display:block;grid-colu
mn:auto;border-left:1px solid #aaa}.population{min-height:180px;gap:2px}.person{print-color-adjust:exa
ct;-webkit-print-color-adjust:exact}.result-number span:first-child{font-size:48px}.counts
strong{font-size:20px}input[type=range]{display:none}.range-row
input{border:0}.scope-note{font-size:10px}.result-panel .ratio-bar,.result-panel
.result-key{display:none}}

```

dist/science.js

```
1 (function(root){
2   'use strict';
3   function calculate(prevalence,sensitivity,specificity,population=1000){
4     for(const [value,label] of
5       [[prevalence,'prevalence'],[sensitivity,'sensitivity'],[specificity,'specificity']])
6       if(!Number.isFinite(value)||value<0||value>100) throw new RangeError(label+' must be a percentage from
7         0 to 100.');
```

```
8     if(!Number.isFinite(population)||population<=0)throw new RangeError('Population must be
9       positive.');
```

```
10     const p=prevalence/100,se=sensitivity/100,sp=specificity/100;
11     const tp=population*p*se,fn=population*p*(1-se),fp=population*(1-p)*(1-sp),tn=population*(1-p)*sp;
12     return {tp,fp,fn,tn,positive:tp+fp,negative:tn+fn,ppv:tp+fp>0?tp/(tp+fp)*100:null,npv:tn+fn>0?tn/(
13       tn+fn)*100:null};
14   }
15   function roundedCounts(counts,total=1000){
16     const names=['tp','fn','fp','tn'],out={},remainders=[];
17     let used=0;
18     for(const name of names){out[name]=Math.floor(counts[name]);used+=out[name];remainders.push([name,
19       counts[name]-out[name]]);}
20     remainders.sort((a,b)=>b[1]-a[1]);
21     for(let i=0;i<total-used;i++)out[remainders[i%names.length][0]]++;
22     return out;
23   }
24   function fromSettings(settings,population=1000){
25     if(!settings||typeof settings!=='object')throw new RangeError('An experiment needs three fictional
26       percentage settings.');
```

```
27     return calculate(settings.prevalence,settings.sensitivity,settings.specificity,population);
28   }
29   function compare(settingsA,settingsB){
30     const a=fromSettings(settingsA),b=fromSettings(settingsB);
31     return {a,b,delta:{tp:b.tp-a.tp,fp:b.fp-a.fp,fn:b.fn-a.fn,tn:b.tn-a.tn,ppv:a.ppv===null||b.ppv===n
32       ull?null:b.ppv-a.ppv}};
33   }
34   function prevalenceCurve(sensitivity,specificity){
35     return
36     Array.from({length:101},(_,i)=>({prevalence:i/2,ppv:calculate(i/2,sensitivity,specificity).ppv}));
37   }
38   // These are conditional rates AMONG first-positive people. Independence is
39   // only one possible way to choose them; repeated results need different rates.
40   function sequential(settings,conditionalDetection,conditionalFalsePositive,population=1000){
41     const first=fromSettings(settings,population);
42     for(const value of
43       [conditionalDetection,conditionalFalsePositive])if(!Number.isFinite(value)||value<0||value>100)throw
44       new RangeError('Conditional rates must be percentages from 0 to 100.');
```

```
45     const tp=first.tp*conditionalDetection/100,fp=first.fp*conditionalFalsePositive/100;
46     return {first,tp,fp,positive:tp+fp,ppv:tp+fp>0?tp/(tp+fp)*100:null,rejectedTrue:first.tp-tp,reject
47       edFalse:first.fp-fp};
48   }
49   function formatNumber(value){
50     if(!Number.isFinite(value))throw new RangeError('Only finite values can be displayed.');
```

```
51     const absolute=Math.abs(value);
52     if(absolute>0&&absolute<1e-7)return value.toExponential(2);
53     const digits=absolute>0&&absolute<1?Math.min(7,Math.ceil(-Math.log10(absolute)))+1):1;
54     return value.toLocaleString('en-US',{maximumFractionDigits:digits});
55   }
56   root.BaseRateScience={calculate,roundedCounts,compare,prevalenceCurve,sequential,formatNumber};
57   if(typeof module!=='undefined')module.exports=root.BaseRateScience;
58 }(typeof window!=='undefined'?window:globalThis));
```

dist/coach.js

```
1 (function(root){
2   'use strict';
3   const content={
4     unavailable:{title:'The AI model did not load.',focus:'Model unavailable',body:'The AI model is
unavailable, so the coach cannot interpret your wording. Your explanation has not been judged. The
mathematical simulator, comparisons and practice remain available.',question:'Reload to try loading
the model, or use the three-case lesson to inspect the arithmetic.',source:'bmj'},
5     balanced:{title:'You are connecting the pieces.',focus:'Prevalence + uncertainty',body:'Your
wording resembles the model\u2019s examples of contextual reasoning. Check that your explanation
connects false positives to the size of the group without the condition.',question:'What changes if
the condition becomes more common but the test stays the same?',source:'bmj'},
6     certainty:{title:'Leave room for uncertainty.',focus:'A result is not a guarantee',body:'The model
detected wording that may treat a screening result as certainty. Positive results can include false
positives; negative results can include missed cases. If you were rejecting certainty, the model may
have misread your wording.',question:'In this experiment, which people show why a positive or negative
result is not a guarantee?',source:'nci'},
7     sensitivity:{title:'Check which group you are counting.',focus:'Sensitivity \u2260 predictive
value',body:'The model detected wording about sensitivity or specificity. Sensitivity starts with
people who have the condition. Positive predictive value starts with people who test positive. Those
denominators are different. If you already made that distinction, the model may have misread your
wording.',question:'To interpret a positive result, should your denominator be all cases or all
positive results?',source:'bmj'},
8     base_rate:{title:'Give prevalence a place in the story.',focus:'The starting population
matters',body:'The model detected wording that may leave out how common the condition is. A large
group without the condition can produce many false positives, even at a low false-positive
rate.',question:'Keep the test settings fixed and change prevalence from 1% to 20%. What changes among
positives?',source:'bmj'},
9     dismissal:{title:'Errors are part of the assessment.',focus:'Benefit and harm need
evidence',body:'The model detected wording that may dismiss all screening because errors occur. False
results matter, but this simulator alone cannot establish a real screening test\u2019s overall
benefits, harms, or suitability.',question:'What evidence would you need beyond these counts to judge
a real screening program?',source:'nci'},
10    uncertain:{title:'Let\u2019s use the counts as our anchor.',focus:'The model is unsure',body:'Your
explanation does not match the model\u2019s limited examples clearly enough. Try stating how
prevalence and false positives affect the group of positive results.',question:'Complete this:
\u201cSome people test positive without the condition, so\u2026\u201d',source:'bmj'},
11    personal:{title:'Keep this experiment fictional.',focus:'Personal results need clinical
context',body:'This lab cannot interpret your personal symptoms, test result, or health risk. Its
numbers describe an invented population. Discuss personal results with an appropriate healthcare
professional.',question:'For this exercise, explain the fictional population shown above
instead.',source:'nci'},
12    empty:{title:'Start with one or two sentences.',focus:'Your reasoning matters',body:'Write an
explanation in your own words, then the coach can look for a reasoning pattern.',question:'Why might a
positive result include a person without the condition?',source:'bmj'}
13  };
14  const sources={bmj:{title:'Gigerenzer \xb7 What are natural frequencies? (BMJ,
2011)',url:'https://www.bmj.com/content/343/bmj.d6386'},nci:{title:'National Cancer Institute \xb7
Cancer Screening
Overview',url:'https://www.cancer.gov/about-cancer/screening/patient-screening-overview-pdq'}};
15  const indexes=new WeakMap();
16  function tokenize(text,model){const stop=new
Set(model.stopWords),words=(text.toLowerCase().replace(/\u2019/g,"").match(/[a-z]+/g)||[]);return
[...words.filter(w=>stop.has(w)),...words.slice(0,-1).map((w,i)=>w+'_'+words[i+1])];}
17  function classify(text,model){
18    if(typeof text!=='string' || text.length>1800)throw new RangeError('Explanation must be text of at
most 1800 characters.');
```

19 if(!text.trim())return {label:'empty',method:'no inference',patterns:[]};

20 if(/(?:\bmys+(:test|result|symptom|diagnosis|blood|scan)|\bhs+(:have|had|got)\s+(?:a\s+)?(?:po
sitive|negative|symptoms|cancer|disease|pain)|\bb(?:diagnose me|should i(?:get|take|stop|start)|do i
have|am i(?:sick|ill)|my chance of))/i.test(text))return {label:'personal',method:'scope
guard',patterns:[]};

21 if(!model)return {label:'unavailable',method:'model unavailable',patterns:[]};

22 if(!indexes.has(label))indexes.set(model,new Map(model.vocabulary.map((v,i)=>[v,i]));)

23 const index=indexes.get(model),feats=tokenize(text,model),known=feats.filter(w=>index.has(w));

24 const scores=model.labels.map(label=>({label,score:model.logPriors[label]+known.reduce((sum,w)=>su
m+model.weights[label][index.get(w)],0)}));

25 const

max=Math.max(...scores.map(s=>s.score)),den=scores.reduce((sum,s)=>sum+Math.exp(s.score-max),0);

26 for(const s of scores)s.vote=Math.exp(s.score-max)/den;

27 scores.sort((a,b)=>b.vote-a.vote);

28 const coverage=known.length/Math.max(1,feats.length),a=model.abstention;

29 const abstain=known.length<a.minKnownFeatures || coverage<a.minCoverage || scores[0].vote<a.minVote || s
cores[0].vote-scores[1].vote<a.minMargin;

30 const first=scores[0],second=scores[1];

31 const patterns=[...new Set(known)].map(word=>({word,diff:model.weights[first.label][index.get(word
)]-model.weights[second.label][index.get(word)]})).filter(x=>x.diff>0).sort((a,b)=>b.diff-a.diff).slic
e(0,4).map(x=>x.word.replace(/_/g,' '));

32 return {label:abstain?'uncertain':first.label,rawLabel:first.label,method:'Multinomial Naive Bayes
,patterns,coverage,vote:first.vote,scores:scores.map(s=>({label:s.label,vote:s.vote})),abstained:abst
ain};

```

33   }
34   function feedback(result,counts){const
item=content[result.label]||content.uncertain,format=root.BaseRateScience?.formatNumber||String;return
  {...item,source:sources[item.source],counts:`In the current fictional experiment:
  ${format(counts.tp)} true positives and ${format(counts.fp)} false positives; ${counts.ppv===null?'the
  positive predictive value is undefined':format(counts.ppv)}% of positive results are expected to be
  true positives'}.`};}
35   root.BaseRateCoach={classify,feedback,tokenize};
36   if(typeof module!=='undefined')module.exports=root.BaseRateCoach;
37 }(typeof window!=='undefined'?window:globalThis);

```

dist/app.js

```
1  'use strict';
2  const $=id=>document.getElementById(id);
3  const settings=[ 'prevalence', 'sensitivity', 'specificity' ];
4  const presets={rare:[1,90,95],common:[20,90,95],specific:[1,90,99]};
5  let positiveOnly=false,current;
6  let lastExplanation=null;
7  function number(n){return BaseRateScience.formatNumber(n);}
8  function update(preserveInput=null){
9    const values=settings.map(id=>Number($ (id).value));
10   current=BaseRateScience.calculate(...values);
11   settings.forEach((id,i)=>{$(id+'-label').textContent=number(values[i]+'%');$(id).setAttribute('aria-
valuertext',number(values[i]+' percent'))});
12   settings.forEach((id,i)=>{const input=$(id+'-number');if(id!==(preserveInput))input.value=values[i];in
put.removeAttribute('aria-invalid')});
13   $('settings-error').textContent='';
14   for(const id of ['tp','fp','fn','tn'])$(id+'-count').textContent=number(current[id]);
15   $('ppv').textContent=current.ppv===null?'2014':number(current.ppv);
16   $('result-explanation').textContent=current.ppv===null?'There are no expected positive results, so
this percentage is undefined.':`About ${number(current.tp)} of ${number(current.positive)} expected
positive results are true positives.`;
17   $('ratio-fill').style.width=(current.ppv??0)+'%';
18   $('formula').textContent=`${number(current.tp)} \xf7 ${number(current.tp)} +
${number(current.fp)} `;
19   $('insight').textContent=current.ppv===null?'A proportion needs a denominator. With no positive
results, there is no positive group to compare.':current.fp>current.tp?'Even a capable test can
produce many false positives when a condition is rare.':`As prevalence rises or false positives fall,
a larger share of positive results can be true positives.`;
20   const counts=BaseRateScience.roundedCounts(current),fragment=document.createDocumentFragment();
21   for(const key of ['tp','fn','fp','tn'])for(let i=0;i<counts[key];i++){const
square=document.createElement('span');square.className='person
'+key+(positiveOnly&&['fn','tn'].includes(key)?'
hidden':'');square.setAttribute('aria-hidden','true');fragment.append(square);}
22   $('population').replaceChildren(fragment);
23   $('population').setAttribute('aria-label`,`Rounded illustration of 1000 people: ${counts.tp} true
positives, ${counts.fp} false positives, ${counts.fn} false negatives, ${counts.tn} true negatives.
${positiveOnly?'Negative results are dimmed.':}``);
24   if(lastExplanation!=null)showFeedback(lastExplanation);
25   window.dispatchEvent(new CustomEvent('baserate:experiment',{detail:{settings:Object.fromEntries(sett
ings.map((id,i)=>[id,values[i]]),counts:{...current}})}));
26 }
27 function setExperiment(values,preserveInput=null){
28   for(const id of settings)if(!Number.isFinite(values?.[id])||values[id]<Number($ (id).min)||values[id]
>Number($ (id).max))throw new RangeError('Fictional '+id+' is outside the available range. ');
29   for(const id of settings)$ (id).value=values[id];
30   $('scenario').value='custom';$('scenario-note').textContent='Explore your own fictional
assumptions.';update(preserveInput);
31 }
32 window.BaseRateApp={getExperiment:()=>({settings:Object.fromEntries(settings.map(id=>[id,Number($ (id).
value)]),counts:{...current}}),setExperiment);
33 function preset(name){if(!presets[name])return;settings.forEach((id,i)=>$(id).value=presets[name][i]);
$('scenario-note').textContent={rare:'A fictional condition affects 1 in 100 people.',common:'The same
fictional test, but the condition affects 1 in 5 people.',specific:'A fictional test produces fewer
false positives.'}[name];update();}
34 settings.forEach(id=>$(id).addEventListener('input',()=>{$( 'scenario').value='custom';$('scenario-note
').textContent='Explore your own fictional assumptions.';update();}));
35 settings.forEach(id=>$(id+'-number').addEventListener('input',()=>{
36   const input=$(id+'-number');
37   if(!input.checkValidity()||!Number.isFinite(input.valueAsNumber)){input.setAttribute('aria-invalid',
'true');$('settings-error').textContent='Enter a value within the labeled range, using the available
step.';return;}
38   input.removeAttribute('aria-invalid');$('settings-error').textContent='';setExperiment({...BaseRateA
pp.getExperiment().settings,[id]:input.valueAsNumber},id);
39 });
40 $('scenario').addEventListener('change',e=>preset(e.target.value));
41 $('reset').addEventListener('click',()=>{$( 'scenario').value='rare';preset('rare')});
42 for(const [id,flag] of [['all-view',false],['positive-view',true],['tree-view',false]])$(id).addEventL
istener('click',()=>{positiveOnly=flag;for(const button of ['all-view','positive-view','tree-view']){$(
button).classList.toggle('active',button===id);$(button).setAttribute('aria-pressed',String(button===
id));}$( 'population-wrap').hidden=id==='tree-view';$( 'frequency-tree').hidden=id==='tree-view';update(
)});
43 $('print').addEventListener('click',()=>window.print());
44 function node(tag,text,className){const element=document.createElement(tag);element.textContent=text;i
f(className)element.className=className;return element;}
45 function showFeedback(text){
46   const result=BaseRateCoach.classify(text>window.BaseRateModel),f=BaseRateCoach.feedback(result,curre
nt),panel=$( 'coach-response');
47   const icon=node('span','\u2733','coach-icon');icon.setAttribute('aria-hidden','true');
48   panel.replaceChildren(icon,node('p',f.focus.toUpperCase(),'small-label'),node('h3',f.title),node('p'
,f.body));
49   if(!['personal','empty'].includes(result.label))panel.append(node('p',f.counts,'current-counts'));
```

```

50   panel.append(node('p',f.question,'coach-question'));
51   if(result.patterns.length&&!result.abstained){const tags=node('div','', 'feedback-tags');for(const
pattern of result.patterns)tags.append(node('span',pattern));panel.append(node('p', 'Patterns that
influenced the model:', 'muted'),tags);}
52   const source=node('div','Read the evidence: ', 'feedback-source'),link=node('a',f.source.title);link.
href=f.source.url;link.target='_blank';link.rel='noopener';source.append(link);panel.append(source);
53   panel.append(node('p','Model feedback is a suggestion, not a grade. A small English classifier can
misread negation or mixed ideas.', 'muted'));
54   if(result.scores){
55     const diagnostics=node('details'),summary=node('summary','Inspect why the model suggested
this');diagnostics.append(summary);
56     diagnostics.append(node('p','Feature coverage: ${number(result.coverage*100)}%.
${result.abstained?'The uncertainty gate withheld the leading label.': 'The leading label passed the
prototype\u2019s uncertainty gate.'}', 'muted'));
57     const bars=node('div','', 'diagnostic-bars'),names={balanced:'Contextual',certainty:'Certainty',sen
sitivity:'Test rates',base_rate:'Prevalence',dismissal:'Dismissal'};
58     for(const score of result.scores){const row=node('div','', 'diagnostic-row'),meter=node('span','', '
diagnostic-meter'),fill=node('span');fill.style.width=score.vote*100+'%';meter.append(fill);row.append
(node('span',names[score.label]),meter,node('span',score.vote>0&&score.vote*100<.1?'<0.1%':number(scor
e.vote*100)+'%'));bars.append(row);}
59     diagnostics.append(bars,node('p','These are relative, uncalibrated model votes. They are not
probabilities of correctness or health outcomes. The feature snippets above are associations, not a
complete explanation.', 'muted'));panel.append(diagnostics);
60   }
61   if(!['personal','empty'].includes(result.label)){const
next=node('a',result.label==='base_rate'? 'Next: compare the starting populations
\u2019: result.label==='sensitivity'? 'Next: practice choosing the denominator \u2019: 'Next: try a
prediction before revealing the counts
\u2019: 'text-link');next.href=result.label==='base_rate'? '#compare': '#practice';panel.append(next);}
62   return {label:result.label,focus:f.focus,counts:{...current},abstained:!!result.abstained};
63 }
64 $('coach-form').addEventListener('submit',e=>{e.preventDefault();lastExplanation=$('explanation').valu
e;showFeedback(lastExplanation)});
65 let exampleIndex=0;
66 const examples=['The test detects 90 percent of cases, so 90 percent of positive results must be real
cases.','A positive result does not prove someone has the condition because false positives can
outnumber true positives when it is rare.','How common the condition is does not matter to a positive
result.'];
67 $('example').addEventListener('click',()=>{ $('explanation').value=examples[exampleIndex++%examples.len
gth];lastExplanation=null; $('explanation').focus();});
68 if(window.BaseRateModel){const m=window.BaseRateModel,e=m.evaluation;
69   $('model-summary').textContent=`A ${m.algorithm} classifier trained on ${m.trainingCount} synthetic
English explanations selects fixed, source-linked educational feedback. Your text is processed on this
device.`;
70   const details=$('model-details');details.replaceChildren(node('p','Version ${m.version}. Word and
adjacent-word features; additive smoothing; five reasoning labels. Training code and dataset are
included in the source package.`,node('p','Development check: ${e.correctCount}/${e.evaluatedCount}
held-out synthetic examples classified correctly; ${e.acceptedCount}/${e.evaluatedCount} received a
label after uncertainty checks. ${e.acceptedCorrect}/${e.acceptedCount} of those labels were
correct.', 'model-metric'),node('p','Known error: one explanation that correctly distinguished
sensitivity from predictive value was misread as confusion. Negation, mixed ideas, unfamiliar wording,
and other languages can fail.`,`),node('p','Uncertainty checks use feature coverage, model vote and
separation between labels. Votes are uncalibrated and are not a probability that the feedback is
correct.`,`),node('p','No real learner study, clinical validation, or measured health benefit. The AI
never computes the simulator\u2019s numbers or generates medical facts. No text is saved or sent by
the app.`,`),node('p','Model and feedback were authored with AI assistance. The synthetic dataset
contains no patient data.`,`));
71 }else{
72   $('model-summary').textContent='The AI model is unavailable. The simulator, comparisons and
mathematical practice still work.`,`;
73   $('model-details').replaceChildren(node('p','Model data did not load. Reload to try again, or
inspect the source package. No AI feedback about your wording can be given until the model is
available.`,`));
74 }
75 if(document.modelContext?.registerTool){
76   const lifecycle=new AbortController();
77   for(const tool of [
78     {name:'configure_fictional_experiment',title:'Configure fictional experiment',description:'Change
the visible educational simulation. No diagnosis or personal risk estimate.',inputSchema:{type:'object',
properties:{prevalence:{type:'number',minimum:0.1,maximum:50},sensitivity:{type:'number',minimum:1,m
aximum:100},specificity:{type:'number',minimum:1,maximum:100},required:settings,additionalProperties:
false},annotations:{readOnlyHint:false},execute(input){for(const id of settings){if(!Number.isFinite(i
nput?.[id])||input[id]<Number($id.min)||input[id]>Number($id.max))throw new Error('Invalid
fictional percentage: '+id);}}for(const id of
settings){$id.value=input[id];$('scenario').value='custom';$('scenario-note').textContent='Explore
your own fictional assumptions.';update();return {settings:Object.fromEntries(settings.map(id=>[id,Num
ber($id.value)]),expectedCounts:{...current}};}},
79     {name:'check_learning_explanation',title:'Check learning explanation',description:'Put an English
explanation into the visible local AI coach and display educational feedback. Text stays on this devic
e.',inputSchema:{type:'object',properties:{explanation:{type:'string',maxLength:1800}},required:['expl
anation'],additionalProperties:false},annotations:{readOnlyHint:false,untrustedContentHint:true},execu
te(input){if(typeof input?.explanation!=='string'||input.explanation.length>1800)throw new
Error('Explanation must be text, at most 1800
characters.`,`);$('explanation').value=input.explanation;lastExplanation=input.explanation;return

```

```

    showFeedback(lastExplanation);}}
80   }){try{Promise.resolve(document.modelContext.registerTool(tool,{signal:lifecycle.signal})).catch(()=>{});}catch{}}
81   window.addEventListener('pagehide',()=>lifecycle.abort(),{once:true});
82 }
83 update();
84 // Fixed, fictional demonstration links; no arbitrary learner text is accepted in URLs.
85 const demo=new URLSearchParams(window.location.search).get('demo');
86 if(demo==='common'){$('#scenario').value='common';preset('common');}
87 if(demo==='coach'){$('#explanation').value=examples[0];lastExplanation=examples[0];showFeedback(lastExplanation);}
88 if(demo==='model')document.querySelector('#evidence_details').open=true;

```

dist/learning.js

```
1 (function(root){
2   'use strict';
3   const science=typeof module!=='undefined'?require('./science.js'):root.BaseRateScience;
4   const cases=Object.freeze([
5     {id:'rare',title:'The rare-condition
6     puzzle',settings:{prevalence:1,sensitivity:90,specificity:95},prompt:'The test detects 90% of cases.
7     What percentage of positive results do you expect to be true positives?',explanation:'The group
8     without the condition is much larger. Five percent of that group produces 49.5 false positives, beside
9     9 true positives. Sensitivity starts with cases; predictive value starts with positives.'},
10    {id:'common',title:'Same test, different
11    population',settings:{prevalence:20,sensitivity:90,specificity:95},prompt:'Only prevalence has
12    changed, from 1% to 20%. Predict the percentage of positive results that are true
13    positives.',explanation:'The test has not improved. The starting population has changed: 180 true
14    positives and 40 false positives. A percentage describing the test alone does not tell the whole
15    story.'},
16    {id:'specific',title:'Fewer false
17    alarms',settings:{prevalence:1,sensitivity:90,specificity:99},prompt:'Prevalence is back to 1%, but
18    specificity is now 99%. Predict the percentage of positive results that are true
19    positives.',explanation:'The smaller false-positive rate gives 9.9 false positives, alongside the same
20    9 true positives. That raises predictive value, but still does not make a positive result a
21    certainty.'}
22  ].map(item=>Object.freeze({...item,settings:Object.freeze(item.settings)})));
23  function gradePrediction(caseId,guess){
24    const item=cases.find(c=>c.id===caseId);
25    if(!item)throw new RangeError('Unknown learning case.');
```

if(!Number.isFinite(guess)||guess<0||guess>100)throw new RangeError('Enter a percentage from 0 to 100.');

```
    const counts=science.calculate(item.settings.prevalence,item.settings.sensitivity,item.settings.specificity);
    const error=Math.abs(counts.ppv-guess);
    return {actual:counts.ppv,error,errorClose:error<=2,count};
  }
  function notebook(answers){
    return {version:1,purpose:'Fictional screening practice; not a validated learning
    assessment',storage:'This export was created on request. The app stores no learner
    data.',attempts:answers.map(answer=>{
26      const
27      item=cases.find(c=>c.id===answer.caseId),result=gradePrediction(answer.caseId,answer.prediction);
28      return {caseId:item.id,settings:{...item.settings},prediction:answer.prediction,actual:result.actual,
29      absoluteErrorPercentagePoints:result.error,denominator:['positives','cases','population'].includes(answer.denominator)?answer.denominator:null};
30    })};
31  }
32  root.BaseRateLearning={cases,gradePrediction,notebook};
33  if(typeof module!=='undefined')module.exports=root.BaseRateLearning;
34  })(typeof window!=='undefined'?window:globalThis);
```

dist/learning-ui.js

```
1 (function(){
2   'use strict';
3   const getId=>document.getElementById(id),cases=BaseRateLearning.cases;
4   let index=0,answers=[],revealed=false,denominatorDone=false;
5   const fmt=value=>BaseRateScience.formatNumber(value);
6   function display(){
7     const item=cases[index];revealed=false;denominatorDone=false;
8     get('practice-progress').textContent=`Case ${index+1} of ${cases.length}`;
9     get('case-title').textContent=item.title;get('case-prompt').textContent=item.prompt;
10    get('case-settings').textContent=`Prevalence ${item.settings.prevalence}% \xb7 Detection
    ${item.settings.sensitivity}% \xb7 Clearance ${item.settings.specificity}%`;
11    get('prediction').value='';get('prediction').disabled=false;
12    get('predict-button').disabled=false;get('prediction-error').textContent='';
13    get('case-reveal').hidden=true;get('practice-next').disabled=true;
14    get('denominator-feedback').textContent='';
15    for(const input of
    document.querySelectorAll('[name="denominator"]')){input.checked=false;input.disabled=false;}
16    get('practice-next').textContent=index===cases.length-1?'See my notebook':'Next case';
17    get('practice-complete').hidden=true;get('practice-case').hidden=false;
18    get('practice-steps').replaceChildren(...cases.map((c,i)=>{
19      const li=document.createElement('li');li.textContent=`0${i+1}
    ${c.title}`;li.className=i===index?'current':i<index?'done':'';
20      if(i===index)li.setAttribute('aria-current','step');return li;
21    }));
22  }
23  get('prediction-form').addEventListener('submit',event=>{
24    event.preventDefault();if(revealed)return;
25    const prediction=get('prediction').valueAsNumber;
26    try{
27      const item=cases[index],result=BaseRateLearning.gradePrediction(item.id,prediction);
28      answers[index]={caseId:item.id,prediction,prediction:null};revealed=true;
29      get('prediction-error').textContent='';
30      get('prediction').disabled=true;get('predict-button').disabled=true;
31      get('case-reveal').hidden=false;
32      get('prediction-result').textContent=`${fmt(result.actual)}% of positive results are true
    positives.`;
33      get('prediction-difference').textContent=`Your first estimate: ${fmt(prediction)}%. Gap:
    ${fmt(result.error)} percentage points. ${result.close?'You were within 2 points.': 'Use the
    denominator below to see the difference.'}`;
34      get('case-equation').textContent=`${fmt(result.counts.tp)} \xf7 (${fmt(result.counts.tp)} +
    ${fmt(result.counts.fp)}) \xd7 100 = ${fmt(result.actual)}%`;
35      get('case-explanation').textContent=item.explanation;
36      get('case-reveal').focus({preventScroll:true});
37    }catch{get('prediction-error').textContent='Enter a number from 0 to 100.';}
38  });
39  get('denominator-form').addEventListener('change',event=>{
40    if(!revealed||denominatorDone||event.target.name!=='denominator')return;
41    denominatorDone=true;answers[index].denominator=event.target.value;
42    get('denominator-feedback').textContent=event.target.value===positives?'Yes. Start with everyone
    who tested positive, then count the true positives inside that group.': 'The denominator is everyone
    who tested positive: true positives plus false positives. Cases alone are the denominator for
    sensitivity.';
43    for(const input of document.querySelectorAll('[name="denominator"]'))input.disabled=true;
44    get('practice-next').disabled=false;
45  });
46  get('practice-next').addEventListener('click',()=>{
47    if(!revealed||!denominatorDone)return;
48    if(index<cases.length-1){index++;display();get('case-title').focus({preventScroll:true});}
49    else{
50      get('practice-case').hidden=true;get('practice-complete').hidden=false;
51      get('notebook-rows').replaceChildren(...answers.map(answer=>{
52        const item=cases.find(c=>c.id===answer.caseId),r=BaseRateLearning.gradePrediction(item.id,answ
    er.prediction),tr=document.createElement('tr');
53        for(const value of [item.title,fmt(answer.prediction)+'%',fmt(r.actual)+'%',fmt(r.error)+'
    pp',answer.denominator===positives?'Positive group':'Review denominator']){const
    cell=document.createElement('td');cell.textContent=value;tr.append(cell);}return tr;
54      }));
55      get('practice-complete').focus({preventScroll:true});
56      for(const li of
    get('practice-steps').children){li.className='done';li.removeAttribute('aria-current');}
57    }
58  });
59  function restart(){index=0;answers=[];display();}
60  get('restart-practice').addEventListener('click',restart);
61  get('download-notebook').addEventListener('click',()=>{
62    const url=URL.createObjectURL(new
    Blob([JSON.stringify(BaseRateLearning.notebook(answers),null,2)],{type:'application/json'}));
63    const link=document.createElement('a');link.href=url;link.download='BaseRate-Learning-Notebook.js
    n';link.click();setTimeout(()=>URL.revokeObjectURL(url),1000);
64  });
65 }
```

```
65 window.BaseRateLesson={notebook:()=>BaseRateLearning.notebook(answers),restart};
66 display();
67 if(new URLSearchParams(window.location.search).get('demo')==='practice'){
68     get('prediction').value=90;get('prediction-form').requestSubmit();
69     get('prediction-difference').textContent='Demonstration estimate: 90%. The expected answer is
15.4%. This is a fictional example, not a learner result.';
70 }
71 }());
```

dist/lab-tools.js

```
1  (function(){
2    'use strict';
3    const get-id=>document.getElementById(id), science=BaseRateScience;
4    const fmt=n=>n===null?'undefined':science.formatNumber(n);
5    const pct=n=>n===null?'Undefined':fmt(n)+'%';
6    const element=(tag, text, css)=>{const
e=document.createElement(tag);if(text!==undefined)e.textContent=text;if(css)e.className=css;return
e;};
7    let pinned={prevalence:1, sensitivity:90, specificity:95};
8    function tree(settings, counts){
9      const root=element('div', '1,000
people', 'tree-root'), branches=element('div', undefined, 'tree-branches');
10     for(const [heading, total, positive, negative, positiveClass, negativeClass] of [
11       ['With the condition', 1000*settings.prevalence/100, counts.tp, counts.fn, 'tp-text', 'fn-text'],
12       ['Without the condition', 1000*(1-settings.prevalence/100), counts.fp, counts.tn, 'fp-text', '']
13     ]){
14       const branch=element('div', undefined, 'tree-branch'), parent=element('div', heading, 'tree-parent');
15       parent.append(element('strong', fmt(total))); branch.append(parent);
16       for(const [label, n, css] of
[[['Positive', positive, positiveClass], ['Negative', negative, negativeClass]]){const
leaf=element('div', undefined, 'tree-leaf '+css); leaf.append(element('span', label), element('b', fmt(n)));
branch.append(leaf);}branches.append(branch);
17     }
18     get('frequency-tree').replaceChildren(root, branches);
19   }
20   function comparison(settings){
21     const pair=science.compare(pinned, settings), rows=[['Prevalence', pct(pinned.prevalence), pct(setting
s.prevalence)], ['Detection / clearance', pct(pinned.sensitivity)+' /
'+pct(pinned.specificity), pct(settings.sensitivity)+' / '+pct(settings.specificity)], ['True
positives', fmt(pair.a.tp), fmt(pair.b.tp)], ['False positives', fmt(pair.a.fp), fmt(pair.b.fp)], ['All
positive results', fmt(pair.a.positive), fmt(pair.b.positive)], ['True among
positives', pct(pair.a.ppv), pct(pair.b.ppv)]];
22     get('compare-rows').replaceChildren(...rows.map(row=>{const
tr=element('tr'); row.forEach((value, i)=>{const
cell=element(i===0?'th':'td', value); if(i===0) cell.scope='row'; tr.append(cell);}); return tr;}));
23     const
sameTest=pinned.sensitivity===settings.sensitivity&&pinned.specificity===settings.specificity;
24     get('compare-insight').textContent=pair.delta.ppv===null?'One experiment has no positive group, so
a percentage-point difference is undefined.': 'B differs by
${pair.delta.ppv}>0?'+' ':'${fmt(pair.delta.ppv)} percentage points.
${sameTest?pinned.prevalence===settings.prevalence?'Both experiments have identical settings.': 'The
test settings are identical; only prevalence differs.'}: 'Prevalence or test settings differ. This
comparison does not isolate a single cause.'`;
25   }
26   const ns='http://www.w3.org/2000/svg';
27   function svgNode(tag, attrs, text){const n=document.createElementNS(ns, tag);for(const [key, value] of
Object.entries(attrs))n.setAttribute(key, value);if(text!==undefined)n.textContent=text;return n;
28   function curve(settings, counts){
29     const svg=get('prevalence-curve'), parts=[svgNode('title', {id:'curve-title'}, 'How prevalence
changes positive predictive value'), svgNode('desc', {id:'curve-description'}, 'Detection
${fmt(settings.sensitivity)} percent, clearance ${fmt(settings.specificity)} percent. Current
prevalence ${fmt(settings.prevalence)} percent gives ${pct(counts.ppv)} true positives among
positives. Equivalent values are in the table.')]
30     const x=p=>60+p/50*540, y=p=>245-p/100*210;
31     for(const value of [0, 25, 50, 75, 100])parts.push(svgNode('line', {x1:60, x2:600, y1:y(value), y2:y(value)
}, class:'curve-grid'), svgNode('text', {x:47, y:y(value)+4, 'text-anchor':'end', class:'curve-axis-label'
}, value+'%'));
32     for(const value of [0, 10, 20, 30, 40, 50])parts.push(svgNode('text', {x:x(value), y:269, 'text-anchor':'m
iddle', class:'curve-axis-label', value+'%'}));
33     parts.push(svgNode('text', {x:60, y:17, class:'curve-axis-label'}, 'True among positive
results'), svgNode('text', {x:330, y:305, 'text-anchor':'middle', class:'curve-axis-label'}, 'Prevalence in
the starting population'));
34     const points=science.prevalenceCurve(settings.sensitivity, settings.specificity); let
path='', connected=false;
35     for(const p of points){if(p.ppv===null){connected=false; continue;}path+=(connected?'L':'M')+x(p.pr
evalence).toFixed(2)+' '+y(p.ppv).toFixed(2)+' '+connected=true;
36     parts.push(svgNode('path', {d:path, class:'curve-line'}));
37     if(counts.ppv!==null){const px=x(settings.prevalence), py=y(counts.ppv); parts.push(svgNode('line', {
x1:px, x2:px, y1:py, y2:245, class:'curve-guide'}, svgNode('circle', {cx:px, cy:py, r:6, class:'curve-dot'}), s
vgNode('text', {x:Math.min(530, Math.max(80, px+13)), y:Math.max(33, py-12), class:'curve-marker-label'}, pct
(counts.ppv)}));
38     svg.replaceChildren(...parts);
39     get('curve-test-label').textContent='Detection ${fmt(settings.sensitivity)}% \xb7 Clearance
${fmt(settings.specificity)}%';
40     get('curve-prevalence').value=settings.prevalence; get('curve-value').textContent=pct(settings.prev
alence);
41     get('curve-table').replaceChildren(...[0, .1, 1, 5, 10, 20, 50].map(p=>{const tr=element('tr'), th=elemen
t('th', pct(p)); th.scope='row'; tr.append(th, element('td', pct(science.calculate(p, settings.sensitivity, s
ettings.specificity).ppv)); return tr;}));
42   }
```

```

42  function sequence(settings){
43    const mode=get('sequence-mode').value;
44    if(mode==='independent'){get('conditional-detection').value=settings.sensitivity;get('conditional-
false').value=Number((100-settings.specificity).toFixed(1));}
45    if(mode==='repeat'){get('conditional-detection').value=100;get('conditional-false').value=100;}
46    const d=Number(get('conditional-detection').value),f=Number(get('conditional-false').value),r=science.sequential(settings,d,f);
47    get('conditional-detection-label').textContent=pct(d);get('conditional-false-label').textContent=pct(f);
48    get('sequence-assumption').textContent=[independent:'Assumption: the two results are independent within each condition group. Real tests can share errors. Independence is invented here, not evidence about a real test.',repeat:'Assumption: every first-positive result is repeated, including every false positive. The second result adds no new information.',custom:'These conditional rates apply only among people positive on the first test. Their values need separate evidence in a real setting.'][mode];
49    get('sequence-ppv').textContent=pct(r.ppv);
50    get('sequence-counts').textContent=r.ppv===null?'No one is expected to be positive twice. The proportion is undefined.':`${fmt(r.tp)} true and ${fmt(r.fp)} false positives remain: ${fmt(r.tp)}\xf7 ${fmt(r.positive)} \xd7 100.`;
51    get('sequence-steps').replaceChildren(element('p','First positives: ${fmt(r.first.tp)} true + ${fmt(r.first.fp)} false'),element('p','Both positive: ${fmt(r.tp)} true + ${fmt(r.fp)} false'),element('p','Second step rejects: ${fmt(r.rejectedTrue)} true + ${fmt(r.rejectedFalse)} false first-positives'));
52  }
53  function render(detail){tree(detail.settings,detail.counts);comparison(detail.settings);curve(detail.settings,detail.counts);sequence(detail.settings);}
54  get('pin-experiment').addEventListener('click',()=>{pinned={...BaseRateApp.getExperiment().settings};get('pin-status').textContent=`A pinned: ${pct(pinned.prevalence)} prevalence, ${pct(pinned.sensitivity)} detection, ${pct(pinned.specificity)} clearance. Changes now affect B only.`;render(BaseRateApp.getExperiment());});
55  get('compare-common').addEventListener('click',()=>{pinned={prevalence:1,sensitivity:90,specificity:95};get('pin-status').textContent=`A: rare condition at 1%. B: same test at 20% prevalence.`;BaseRateApp.setExperiment({prevalence:20,sensitivity:90,specificity:95});});
56  get('curve-prevalence').addEventListener('input',()=>BaseRateApp.setExperiment({...BaseRateApp.getExperiment().settings,prevalence:Number(get('curve-prevalence').value)}));
57  get('sequence-mode').addEventListener('change',()=>sequence(BaseRateApp.getExperiment().settings));
58  for(const id of ['conditional-detection','conditional-false'])get(id).addEventListener('input',()=>{get('sequence-mode').value='custom';sequence(BaseRateApp.getExperiment().settings)});
59  get('share-experiment').addEventListener('click',async()=>{
60    const url=new URL(window.location.href);url.search='';url.hash='lab';
61    const s=BaseRateApp.getExperiment().settings;url.searchParams.set('p',s.prevalence);url.searchParams.set('se',s.sensitivity);url.searchParams.set('sp',s.specificity);
62    try{await navigator.clipboard.writeText(url.href);get('link-message').textContent='Copied. This link contains only fictional rate settings.';}catch{get('link-message').textContent='Copy this settings link: ${url.href}`;}}
63  });
64  window.addEventListener('baserate:experiment',event=>render(event.detail));
65  const params=new URLSearchParams(window.location.search),keys=['p','se','sp'];
66  if(keys.some(key=>params.has(key))){
67    const valid=keys.every(key=>params.getAll(key).length===1&&/^(?:\d+(?:\.\d+)?|\.\d+)$/.test(params.get(key)));
68    try{if(!valid)throw new Error('Invalid settings');BaseRateApp.setExperiment({prevalence:Number(params.get('p')),sensitivity:Number(params.get('se')),specificity:Number(params.get('sp'))});get('link-message').textContent='Loaded fictional settings from this link.';}catch{get('link-message').textContent='This settings link is incomplete or outside the available ranges. The default experiment is shown.';}}
69  }
70  render(BaseRateApp.getExperiment());
71  const demo=new URLSearchParams(window.location.search).get('demo');
72  if(demo==='compare')BaseRateApp.setExperiment({prevalence:20,sensitivity:90,specificity:95});
73  if(demo==='sequence')get('sequence').open=true;
74  })();

```

model/train.py

```
1  """Reproducible, dependency-free educational text-model training. No clinical data."""
2  import collections, hashlib, json, math, pathlib, re
3  ROOT = pathlib.Path(__file__).resolve().parents[1]
4  data = json.loads((ROOT / 'model/dataset.json').read_text())
5  STOP = {'a', 'an', 'the', 'is', 'are', 'am', 'be', 'being', 'been', 'of', 'to', 'in', 'and', 'or', 'it', 'its', 'if', 'for', 'as', 'at', 'i', 'we', 'you', 'they', 'their', 'there', 'this', 'that', 'these', 'those', 'with', 'by', 'from', 'on', 'us', 'our', 'my', 'has', 'have', 'had', 'was', 'were', 'can', 'could', 'would', 'should'}
6  def features(text):
7      words = re.findall(r"[a-z]+", text.lower().replace('\u2019', ''))
8      return [w for w in words if w not in STOP] + [words[i]+'_'+words[i+1] for i in range(len(words)-1)]
9  labels = list(data['train'])
10 counts = {label: collections.Counter() for label in labels}
11 vocabulary = set()
12 for label, examples in data['train'].items():
13     for example in examples:
14         counts[label].update(features(example))
15         vocabulary.update(features(example))
16 vocabulary = sorted(vocabulary)
17 alpha = 0.7
18 weights = {label:
19     [math.log((counts[label][word]+alpha)/(sum(counts[label].values()+alpha*len(vocabulary)))) for word in vocabulary] for label in labels}
19 model = {'version':'1.0.0', 'algorithm':'Multinomial Naive Bayes', 'labels':labels, 'vocabulary':vocabulary, 'weights':weights, 'logPriors':{l:math.log(len(data['train'][l])/sum(map(len,data['train'].values()))) for l in labels}, 'stopWords':sorted(STOP), 'trainingCount':sum(map(len,data['train'].values())), 'holdoutCount':sum(map(len,data['holdout'].values())), 'datasetSha256':hashlib.sha256((ROOT/'model/dataset.json').read_bytes()).hexdigest(), 'abstention':{'minKnownFeatures':4, 'minCoverage':0.18, 'minVote':0.65, 'minMargin':0.2}, 'warning':'Scores are uncalibrated model votes, not probabilities of correctness. Synthetic English evaluation is not evidence of clinical or learner benefit.'}
20 index = {word:i for i,word in enumerate(vocabulary)}
21 def predict(text):
22     feats=features(text)
23     known=[index[w] for w in feats if w in index]
24     scores={label:model['logPriors'][label]+sum(weights[label][i] for i in known) for label in labels}
25     maximum=max(scores.values())
26     exp={k:math.exp(v-maximum) for k,v in scores.items()}
27     votes={k:v/sum(exp.values()) for k,v in exp.items()}
28     ordered=sorted(votes, key=votes.get, reverse=True)
29     a=model['abstention']
30     abstain=len(known)<a['minKnownFeatures'] or len(known)/max(1, len(feats))<a['minCoverage'] or votes[ordered[0]]<a['minVote'] or votes[ordered[0]]-votes[ordered[1]]<a['minMargin']
31     return ordered[0], votes[ordered[0]], abstain
32 confusion={l:{p:0 for p in labels} for l in labels}
33 rows=[]
34 for actual, examples in data['holdout'].items():
35     for text in examples:
36         predicted, vote, abstain=predict(text)
37         confusion[actual][predicted]+=1
38         rows.append({'text':text, 'actual':actual, 'predicted':predicted, 'vote':round(vote,4), 'abstained':abstain})
39 correct=sum(r['actual']==r['predicted'] for r in rows)
40 accepted=[r for r in rows if not r['abstained']]
41 metrics={'evaluatedCount':len(rows), 'correctCount':correct, 'rawAccuracy':correct/len(rows), 'acceptedCount':len(accepted), 'acceptedCorrect':sum(r['actual']==r['predicted'] for r in accepted), 'coverage':len(accepted)/len(rows), 'confusionMatrix':confusion, 'limits':'Small, author-created English holdout. Training and holdout share a fictional task; no external, multilingual, clinical, or educational-outcome validation.'}
42 model['evaluation']=metrics
43 (ROOT/'dist/model.js').write_text('window.BaseRateModel = '+json.dumps(model, separators=(',', ':'))+';\n')
44 (ROOT/'model/evaluation.json').write_text(json.dumps({'metrics':metrics, 'predictions':rows}, indent=2)+'\n')
45 print(json.dumps(metrics, indent=2))
```

model/dataset.json

```
1 {
2   "description": "Author-created fictional English learner explanations. Labels describe educational
   reasoning, never a health status. Held-out sentences were authored separately; no real learners or
   patient data are included.",
3   "train": {
4     "balanced": [
5       "A positive result does not prove the condition because false positives happen.",
6       "The chance after a positive result depends on how common the condition is.",
7       "We need to count true positives and false positives among everyone who tests positive.",
8       "A rare condition means many people without it may test positive.",
9       "Sensitivity is not the same as the chance of the condition after a positive test.",
10      "Prevalence matters because the group without the condition can be much larger.",
11      "A negative result does not completely rule out the condition because cases can be missed.",
12      "Both false positives and false negatives are possible in a screening test.",
13      "The proportion with the condition among positives is true positives divided by all positives.",
14      "A result is uncertain and has to be interpreted in context.",
15      "Most positives could be false when the condition is uncommon.",
16      "Even a good test cannot make a positive result a guaranteed diagnosis.",
17      "The same sensitivity can give different predictive values in different populations.",
18      "The result depends on prevalence and the number of people incorrectly flagged.",
19      "Testing positive is not a certainty; some positive people do not have the condition.",
20      "False positives can outnumber true positives when there are few actual cases.",
21      "A positive is a clue rather than proof, since the test can be wrong.",
22      "Changing how common the condition is changes the meaning of a positive result.",
23      "To understand a positive we compare detected cases with all positive results.",
24      "A negative test can miss cases, so it is not an absolute guarantee."
25    ],
26    "certainty": [
27      "A positive result proves you have the condition.",
28      "Everyone with a positive test definitely has the disease.",
29      "Testing positive means a guaranteed diagnosis.",
30      "The person must be ill because their test is positive.",
31      "A positive result is certain proof of the condition.",
32      "There is no uncertainty after a positive test.",
33      "A negative result proves you cannot have the condition.",
34      "If the result is negative it is impossible to have the disease.",
35      "Everyone who tests negative is definitely healthy.",
36      "A negative test guarantees there are no cases missed.",
37      "The test is positive so there is a hundred percent chance of disease.",
38      "Positive tests always mean actual disease.",
39      "The result is conclusive and there cannot be any mistakes.",
40      "A positive is a definite diagnosis with no need for context.",
41      "You always have the condition if the screening says positive.",
42      "A negative rules out disease completely.",
43      "Screening gives absolute certainty about who is ill.",
44      "The result tells us for sure whether the condition is present.",
45      "No positive person could be free of the disease.",
46      "Nobody with a negative result can have the condition."
47    ],
48    "sensitivity": [
49      "The sensitivity is ninety percent so a positive person has a ninety percent chance of
   disease.",
50      "Sensitivity tells us the chance of being ill after a positive result.",
51      "Ninety percent sensitivity means ninety percent of positive results are true.",
52      "Positive predictive value and sensitivity are the same number.",
53      "The detection rate is the probability a positive person has the condition.",
54      "A test that detects ninety percent of cases means only ten percent of positives are false.",
55      "The sensitivity directly answers how likely disease is after testing positive.",
56      "I use the sensitivity as the probability of disease in the positive group.",
57      "A ninety percent detection rate gives every positive person ninety percent risk.",
58      "Since the sensitivity is high most positive people have the condition regardless of
   prevalence.",
59      "Specificity is the percentage of positive results that are correct.",
60      "The specificity tells how many of the positive group are really ill.",
61      "Ninety five percent specificity means ninety five percent of positives have the disease.",
62      "The test accuracy is simply the chance of disease if positive.",
63      "The sensitivity percentage can be read as the probability of a correct positive.",
64      "True positives over all positives is the definition of sensitivity.",
65      "Sensitivity and the proportion ill among positive tests are identical.",
66      "A detection rate of ninety percent means nine out of ten positive tests are actual cases.",
67      "You can use specificity to find the disease chance among people testing positive.",
68      "The probability after a positive test is equal to the detection rate."
69    ],
70    "base_rate": [
71      "How common the condition is does not matter to the result.",
72      "Prevalence is irrelevant if the test is accurate.",
73      "The number of people with the condition makes no difference.",
74      "A positive result means the same thing in every population.",
75      "We can ignore the base rate because the test performs well.",
76      "The chance after a positive is unchanged when prevalence changes."
```

```

77     "It does not matter whether the condition is rare or common.",
78     "All groups have the same predictive value if they use the same test.",
79     "We only need sensitivity and specificity, not prevalence.",
80     "The group size without disease is irrelevant to the positive results.",
81     "Changing the population cannot change the meaning of a positive result.",
82     "The frequency of the condition has nothing to do with interpreting tests.",
83     "Rare and common conditions produce the same probability after a positive.",
84     "Prevalence should be left out of the calculation.",
85     "The base rate has no effect on positive predictive value.",
86     "We do not need to know how widespread the condition is.",
87     "An accurate test makes the prevalence unimportant.",
88     "Testing a different group will not change the proportion of true positives.",
89     "How many people start with disease is not needed for a positive prediction.",
90     "I would ignore prevalence and focus only on the test performance."
91 ],
92 "dismissal": [
93     "False positives mean screening tests are completely useless.",
94     "If there are false positives we should never use screening.",
95     "Any false negative means the test has no value at all.",
96     "A test with mistakes is worthless and should be thrown away.",
97     "Screening is pointless because it cannot be perfect.",
98     "If most positives are false all screening is a waste of time.",
99     "False positives prove no one should ever be screened.",
100    "An imperfect test cannot help anybody.",
101    "We should reject every screening test that sometimes gives false results.",
102    "All screening tests are bad because false alarms happen.",
103    "A false positive makes the whole test useless.",
104    "There is no benefit to screening if any cases are missed.",
105    "The only worthwhile test is a perfect test with no errors.",
106    "A test that misses even one case should never be used.",
107    "These errors show screening has no purpose.",
108    "A lot of false alarms means screening is always harmful.",
109    "I would dismiss screening altogether because false positives exist.",
110    "Any mistake makes the test totally pointless.",
111    "Tests with false positives cannot ever be useful.",
112    "An imperfect screen is a waste and has zero benefit."
113 ],
114 },
115 "holdout": {
116     "balanced": [
117         "Positive results can include people without the condition, so I compare true positives with all
positives.",
118         "If a condition is rare, the large healthy group can create lots of false positives.",
119         "A positive screening result is not a diagnosis because the test sometimes flags non cases.",
120         "A negative result is not certain because false negatives can occur.",
121         "How common the condition is affects the proportion of real cases among positive results.",
122         "Sensitivity measures detected cases rather than the probability of disease among positives."
123     ],
124     "certainty": [
125         "If someone screens positive they must have the condition for sure.",
126         "A negative result is an absolute guarantee of no disease.",
127         "The screen confirms the disease with complete certainty.",
128         "Positive means definitely ill and negative means definitely well.",
129         "Every positive result is conclusive proof of disease.",
130         "It is impossible for somebody with a negative screen to be ill."
131     ],
132     "sensitivity": [
133         "The test detects ninety percent of cases, therefore ninety percent of positive people are
sick.",
134         "I think sensitivity is the chance a positive test is a real case.",
135         "The ninety percent sensitivity is also the positive predictive value.",
136         "Specificity is how likely somebody is to have the condition given a positive.",
137         "Sensitivity means the proportion with disease in the positive group.",
138         "Ninety percent case detection tells us nine out of ten positives have disease."
139     ],
140     "base_rate": [
141         "The disease frequency is irrelevant to interpreting a positive test.",
142         "I can leave prevalence out because sensitivity and specificity are enough.",
143         "It makes no difference if the disease is uncommon or widespread.",
144         "The same test means the same probability in any group.",
145         "The base rate does not affect the fraction of true positives.",
146         "There is no reason to consider how common the condition is."
147     ],
148     "dismissal": [
149         "Because false positives occur, screening should never be used at all.",
150         "A screening test that misses a case is entirely worthless.",
151         "False alarms mean the test has zero value.",
152         "An imperfect test is always a waste of effort.",
153         "Any false result proves screening is pointless.",
154         "If tests are not perfect there cannot be any screening benefit."
155     ]
156 }
157 }

```

model/evaluation.json

```
1 {
2   "metrics": {
3     "evaluatedCount": 30,
4     "correctCount": 29,
5     "rawAccuracy": 0.9666666666666667,
6     "acceptedCount": 28,
7     "acceptedCorrect": 27,
8     "coverage": 0.9333333333333333,
9     "confusionMatrix": {
10      "balanced": {
11        "balanced": 5,
12        "certainty": 0,
13        "sensitivity": 1,
14        "base_rate": 0,
15        "dismissal": 0
16      },
17      "certainty": {
18        "balanced": 0,
19        "certainty": 6,
20        "sensitivity": 0,
21        "base_rate": 0,
22        "dismissal": 0
23      },
24      "sensitivity": {
25        "balanced": 0,
26        "certainty": 0,
27        "sensitivity": 6,
28        "base_rate": 0,
29        "dismissal": 0
30      },
31      "base_rate": {
32        "balanced": 0,
33        "certainty": 0,
34        "sensitivity": 0,
35        "base_rate": 6,
36        "dismissal": 0
37      },
38      "dismissal": {
39        "balanced": 0,
40        "certainty": 0,
41        "sensitivity": 0,
42        "base_rate": 0,
43        "dismissal": 6
44      }
45    },
46     "limits": "Small, author-created English holdout. Training and holdout share a fictional task; no
external, multilingual, clinical, or educational-outcome validation."
47   },
48   "predictions": [
49     {
50       "text": "Positive results can include people without the condition, so I compare true positives
with all positives.",
51       "actual": "balanced",
52       "predicted": "balanced",
53       "vote": 0.9999,
54       "abstained": false
55     },
56     {
57       "text": "If a condition is rare, the large healthy group can create lots of false positives.",
58       "actual": "balanced",
59       "predicted": "balanced",
60       "vote": 0.9628,
61       "abstained": false
62     },
63     {
64       "text": "A positive screening result is not a diagnosis because the test sometimes flags non
cases.",
65       "actual": "balanced",
66       "predicted": "balanced",
67       "vote": 0.9848,
68       "abstained": false
69     },
70     {
71       "text": "A negative result is not certain because false negatives can occur.",
72       "actual": "balanced",
73       "predicted": "balanced",
74       "vote": 0.9855,
75       "abstained": false
76     },
77   ]
}
```

```

78     "text": "How common the condition is affects the proportion of real cases among positive
results.",
79     "actual": "balanced",
80     "predicted": "balanced",
81     "vote": 0.955,
82     "abstained": false
83 },
84 {
85     "text": "Sensitivity measures detected cases rather than the probability of disease among
positives.",
86     "actual": "balanced",
87     "predicted": "sensitivity",
88     "vote": 0.9808,
89     "abstained": false
90 },
91 {
92     "text": "If someone screens positive they must have the condition for sure.",
93     "actual": "certainty",
94     "predicted": "certainty",
95     "vote": 0.9712,
96     "abstained": false
97 },
98 {
99     "text": "A negative result is an absolute guarantee of no disease.",
100    "actual": "certainty",
101    "predicted": "certainty",
102    "vote": 0.9967,
103    "abstained": false
104 },
105 {
106    "text": "The screen confirms the disease with complete certainty.",
107    "actual": "certainty",
108    "predicted": "certainty",
109    "vote": 0.8101,
110    "abstained": false
111 },
112 {
113    "text": "Positive means definitely ill and negative means definitely well.",
114    "actual": "certainty",
115    "predicted": "certainty",
116    "vote": 0.9847,
117    "abstained": false
118 },
119 {
120    "text": "Every positive result is conclusive proof of disease.",
121    "actual": "certainty",
122    "predicted": "certainty",
123    "vote": 0.996,
124    "abstained": false
125 },
126 {
127    "text": "It is impossible for somebody with a negative screen to be ill.",
128    "actual": "certainty",
129    "predicted": "certainty",
130    "vote": 0.9991,
131    "abstained": false
132 },
133 {
134    "text": "The test detects ninety percent of cases, therefore ninety percent of positive people
are sick.",
135    "actual": "sensitivity",
136    "predicted": "sensitivity",
137    "vote": 1.0,
138    "abstained": false
139 },
140 {
141    "text": "I think sensitivity is the chance a positive test is a real case.",
142    "actual": "sensitivity",
143    "predicted": "sensitivity",
144    "vote": 0.9525,
145    "abstained": false
146 },
147 {
148    "text": "The ninety percent sensitivity is also the positive predictive value.",
149    "actual": "sensitivity",
150    "predicted": "sensitivity",
151    "vote": 1.0,
152    "abstained": false
153 },
154 {
155    "text": "Specificity is how likely somebody is to have the condition given a positive.",
156    "actual": "sensitivity",
157    "predicted": "sensitivity",
158    "vote": 0.6772,

```

```

159     "abstained": false
160 },
161 {
162     "text": "Sensitivity means the proportion with disease in the positive group.",
163     "actual": "sensitivity",
164     "predicted": "sensitivity",
165     "vote": 0.9981,
166     "abstained": false
167 },
168 {
169     "text": "Ninety percent case detection tells us nine out of ten positives have disease.",
170     "actual": "sensitivity",
171     "predicted": "sensitivity",
172     "vote": 1.0,
173     "abstained": false
174 },
175 {
176     "text": "The disease frequency is irrelevant to interpreting a positive test.",
177     "actual": "base_rate",
178     "predicted": "base_rate",
179     "vote": 0.5632,
180     "abstained": true
181 },
182 {
183     "text": "I can leave prevalence out because sensitivity and specificity are enough.",
184     "actual": "base_rate",
185     "predicted": "base_rate",
186     "vote": 0.6504,
187     "abstained": false
188 },
189 {
190     "text": "It makes no difference if the disease is uncommon or widespread.",
191     "actual": "base_rate",
192     "predicted": "base_rate",
193     "vote": 0.869,
194     "abstained": false
195 },
196 {
197     "text": "The same test means the same probability in any group.",
198     "actual": "base_rate",
199     "predicted": "base_rate",
200     "vote": 0.9962,
201     "abstained": false
202 },
203 {
204     "text": "The base rate does not affect the fraction of true positives.",
205     "actual": "base_rate",
206     "predicted": "base_rate",
207     "vote": 0.9767,
208     "abstained": false
209 },
210 {
211     "text": "There is no reason to consider how common the condition is.",
212     "actual": "base_rate",
213     "predicted": "base_rate",
214     "vote": 0.6138,
215     "abstained": true
216 },
217 {
218     "text": "Because false positives occur, screening should never be used at all.",
219     "actual": "dismissal",
220     "predicted": "dismissal",
221     "vote": 1.0,
222     "abstained": false
223 },
224 {
225     "text": "A screening test that misses a case is entirely worthless.",
226     "actual": "dismissal",
227     "predicted": "dismissal",
228     "vote": 0.9985,
229     "abstained": false
230 },
231 {
232     "text": "False alarms mean the test has zero value.",
233     "actual": "dismissal",
234     "predicted": "dismissal",
235     "vote": 0.9997,
236     "abstained": false
237 },
238 {
239     "text": "An imperfect test is always a waste of effort.",
240     "actual": "dismissal",
241     "predicted": "dismissal",
242     "vote": 0.9993,

```

```
243     "abstained": false
244   },
245   {
246     "text": "Any false result proves screening is pointless.",
247     "actual": "dismissal",
248     "predicted": "dismissal",
249     "vote": 0.9869,
250     "abstained": false
251   },
252   {
253     "text": "If tests are not perfect there cannot be any screening benefit.",
254     "actual": "dismissal",
255     "predicted": "dismissal",
256     "vote": 0.991,
257     "abstained": false
258   }
259 ]
260 }
```

tests/verify.cjs

```
1  const assert=require('node:assert/strict');
2  const fs=require('node:fs');
3  const vm=require('node:vm');
4  const path=require('node:path');
5  const root=path.resolve(__dirname, '..');
6  const science=require('../dist/science.js');
7  const coach=require('../dist/coach.js');
8  const context={window:{}};
9  vm.runInNewContext(fs.readFileSync(path.join(root, 'dist/model.js'), 'utf8'), context);
10 const model=context.window.BaseRateModel;
11 function close(a,b){assert.ok(Math.abs(a-b)<1e-9, `${a} != ${b}`);}
12 const rare=science.calculate(1,90,95);
13 close(rare.tp,9);close(rare.fp,49.5);close(rare.fn,1);close(rare.tn,940.5);close(rare.ppv,100*9/58.5);
14 close(science.calculate(20,90,95).ppv,100*180/220);
15 close(science.calculate(1,90,99).ppv,100*9/18.9);
16 assert.equal(science.calculate(0,90,100).ppv,null);
17 assert.equal(science.calculate(100,100,0).npv,null);
18 assert.equal(science.calculate(1,100,100).ppv,100);
19 assert.throws(()=>science.calculate(NaN,90,95),RangeError);
20 assert.throws(()=>science.calculate(101,90,95),RangeError);
21 assert.throws(()=>science.calculate(1,90,95,0),RangeError);
22 let combinations=0;
23 for(const p of [0,.1,1,7.3,20,50,99.9,100])for(const se of [0,1,10,63,90,99.9,100])for(const sp of
[0,1,50,95,99.9,100]){
24   const r=science.calculate(p,se,sp),rounded=science.roundedCounts(r);
25   close(r.tp+r.fp+r.fn+r.tn,1000);
26   assert.equal(Object.values(rounded).reduce((a,b)=>a+b,0),1000);
27   for(const key of
['tp','fn','fp','tn']){assert.ok(r[key]>=0);assert.ok(Math.abs(rounded[key]-r[key])<=1.0000001);}
28   for(const value of [r.ppv,r.npv])assert.ok(value===null||(value>=0&&value<=100));
29   close(r.ppv??0,science.calculate(p,se,sp,10000).ppv??0);
30   combinations++;
31 }
32 assert.equal(coach.classify('',model).label,'empty');
33 assert.equal(coach.classify('My test result came back positive, do I have
cancer?',model).label,'personal');
34 assert.equal(coach.classify('\u4f60\u597d\u4e16\u754c',model).label,'uncertain');
35 assert.equal(coach.classify('banana bicycle orchestra',model).label,'uncertain');
36 assert.throws(()=>coach.classify('x'.repeat(1801),model),RangeError);
37 assert.equal(coach.classify('A positive result does not prove the condition because false positives
happen.',model).label,'balanced');
38 assert.equal(coach.classify('A positive result proves you have the
condition.',model).label,'certainty');
39 const expected=JSON.parse(fs.readFileSync(path.join(root, 'model/evaluation.json'), 'utf8'));
40 for(const row of expected.predictions){const actual=coach.classify(row.text,model);assert.equal(actual
.rawLabel,row.predicted);assert.equal(actual.abstained,row.abstained);}
41 const dataset=JSON.parse(fs.readFileSync(path.join(root, 'model/dataset.json'), 'utf8'));
42 const train=new Set(Object.values(dataset.train).flat());
43 for(const text of Object.values(dataset.holdout).flat())assert.ok(!train.has(text),'Holdout sentence
occurs in training');
44 const sourceFiles=['index.html','style.css','science.js','model.js','coach.js','app.js'];
45 for(const file of sourceFiles)assert.ok(fs.statSync(path.join(root, 'dist', file)).size>0);
46 console.log(JSON.stringify({result:'PASS',scienceCombinations:combinations,modelParityCases:expected.p
redictions.length,checks:['Bayes examples','edge denominators','count
conservation','rounding','invalid inputs','scope guard','out of vocabulary abstention','negation
example','train/holdout disjointness','browser/Python model parity']}));
```

tests/extensions.cjs

```
1  const assert=require('node:assert/strict');
2  const science=require('../dist/science.js');
3  const rare={prevalence:1,sensitivity:90,specificity:95};
4  const common={...rare,prevalence:20};
5  const close=(a,b)=>assert.ok(Math.abs(a-b)<1e-8,`${a} != ${b}`);
6  const pair=science.compare(rare,common);
7  close(pair.a.ppv,9/58.5*100);
8  close(pair.b.ppv,180/220*100);
9  close(pair.delta.ppv,180/220*100-9/58.5*100);
10 close(pair.delta.fp,-9.5);close(pair.delta.tp,171);
11 assert.deepEqual(rare,{prevalence:1,sensitivity:90,specificity:95});
12 assert.equal(science.compare({prevalence:0,sensitivity:90,specificity:100},rare).delta.ppv,null);
13 let curvePoints=0;
14 for(const se of [0,1,50,90,100])for(const sp of [0,1,50,95,100]){
15   const curve=science.prevalenceCurve(se,sp);
16   assert.equal(curve.length,101);assert.equal(curve[0].prevalence,0);assert.equal(curve[100].prevalenc
e,50);
17   let previous=0;
18   for(const point of curve){
19     close(point.prevalence*2,Math.round(point.prevalence*2));
20     assert.equal(point.ppv,science.calculate(point.prevalence,se,sp).ppv);
21     if(point.ppv!==null){assert.ok(point.ppv+1e-8>=previous);previous=point.ppv;}
22     curvePoints++;
23   }
24 }
25 const independent=science.sequential(rare,90,5);
26 close(independent.tp,8.1);close(independent.fp,2.475);
27 close(independent.ppv,8.1/(8.1+2.475)*100);
28 close(independent.rejectedTrue,.9);close(independent.rejectedFalse,47.025);
29 const repeated=science.sequential(rare,100,100);
30 close(repeated.ppv,science.calculate(1,90,95).ppv);
31 close(repeated.tp,9);close(repeated.fp,49.5);
32 assert.equal(science.sequential(rare,0,0).ppv,null);
33 assert.equal(science.sequential(rare,100,0).ppv,100);
34 let sequenceCases=0;
35 for(const p of [0,.1,1,20,100])for(const se of [0,90,100])for(const sp of [0,95,100])for(const d of
[0,90,100])for(const f of [0,5,100]){
36   const r=science.sequential({prevalence:p,sensitivity:se,specificity:sp},d,f);
37   close(r.tp+r.rejectedTrue,r.first.tp);close(r.fp+r.rejectedFalse,r.first.fp);
38   assert.ok(r.tp>=0&&r.fp>=0&&r.rejectedTrue>=-1e-10&&r.rejectedFalse>=-1e-10);
39   assert.ok(r.ppv===null||(r.ppv>=0&&r.ppv<=100));sequenceCases++;
40 }
41 for(const bad of [NaN,Infinity,-1,101]){
42   assert.throws(()=>science.prevalenceCurve(bad,90),RangeError);
43   assert.throws(()=>science.sequential(rare,bad,5),RangeError);
44   assert.throws(()=>science.sequential(rare,90,bad),RangeError);
45   assert.throws(()=>science.compare({...rare,prevalence:bad},common),RangeError);
46 }
47 console.log(JSON.stringify({result:'PASS',curvePoints,sequenceCases,checks:['comparison
units','immutable inputs','monotonic prevalence curve','conditional repeated tests','undefined
proportions','conservation','invalid rates']}));
```

tests/learning.cjs

```
1  const assert=require('node:assert/strict');
2  const learning=require('../dist/learning.js');
3  const close=(a,b)=>assert.ok(Math.abs(a-b)<1e-9);
4  assert.equal(learning.cases.length,3);
5  assert.ok(Object.isFrozen(learning.cases[0].settings));
6  close(learning.gradePrediction('rare',90).actual,9/58.5*100);
7  close(learning.gradePrediction('rare',90).error,90-9/58.5*100);
8  close(learning.gradePrediction('common',80).actual,180/220*100);
9  close(learning.gradePrediction('specific',50).actual,9/18.9*100);
10 assert.equal(learning.gradePrediction('rare',15.4).close,true);
11 assert.equal(learning.gradePrediction('rare',90).close,false);
12 for(const guess of [0,100])assert.ok(Number.isFinite(learning.gradePrediction('rare',guess).error));
13 for(const guess of ['',null,undefined,NaN,Infinity,-.1,100.1,'15'])assert.throws(()=>learning.gradePrediction('rare',guess),RangeError);
14 assert.throws(()=>learning.gradePrediction('missing',50),RangeError);
15 const answers=[{caseId:'rare',prediction:90,denominator:'cases',explanation:'private learner words',arbitrary:'secret'}];
16 const notebook=learning.notebook(answers);
17 assert.equal(notebook.attempts.length,1);assert.equal(notebook.attempts[0].prediction,90);
18 assert.equal(notebook.attempts[0].denominator,'cases');
19 assert.equal(JSON.stringify(notebook).includes('private learner words'),false);
20 assert.equal(JSON.stringify(notebook).includes('secret'),false);
21 assert.equal(notebook.attempts[0].settings.prevalence,1);
22 assert.equal(learning.notebook([]).attempts.length,0);
23 assert.deepEqual(answers,[{caseId:'rare',prediction:90,denominator:'cases',explanation:'private learner words',arbitrary:'secret'}]);
24 console.log(JSON.stringify({result:'PASS',checks:['three case calculations','first guess error','immutable content','invalid guesses','text-free notebook','empty notebook']}));
```

tests/display.cjs

```
1  const assert=require('node:assert/strict');
2  const science=require('../dist/science.js');
3  const coach=require('../dist/coach.js');
4  const result=science.sequential({prevalence:.1,sensitivity:1,specificity:100},1,0);
5  assert.equal(result.ppv,100);
6  assert.equal(science.formatNumber(result.tp),'0.0001');
7  assert.equal(science.formatNumber(result.positive),'0.0001');
8  assert.equal(science.formatNumber(result.first.tp),'0.01');
9  assert.equal(science.formatNumber(49.5),'49.5');
10 assert.equal(science.formatNumber(940.5),'940.5');
11 assert.equal(science.formatNumber(0),'0');
12 for(const value of [.000000000001,.0001,.001,.01,.1,-.0001])assert.notEqual(Number(science.formatNumber
(value).replace(/,/g,'')),0);
13 const unavailable=coach.classify('False positives can outnumber true positives.',null);
14 assert.equal(unavailable.label,'unavailable');
15 assert.match(coach.feedback(unavailable,result.first).body,/model.*unavailable/i);
16 assert.match(coach.feedback(unavailable,result.first).counts,/0\.01 true positives/);
17 assert.match(coach.feedback(unavailable,result).counts,/0\.0001 true positives/);
18 console.log(JSON.stringify({result:'PASS',checks:['nonzero count precision','defined repeat
denominator','very small display values','missing model message']}));
```

dist/model.js

```
1 window.BaseRateModel = {
2   "version": "1.0.0",
3   "algorithm": "Multinomial Naive Bayes",
4   "labels": [
5     "balanced",
6     "certainty",
7     "sensitivity",
8     "base_rate",
9     "dismissal"
10  ],
11  "vocabulary": [
12    "a_certainty",
13    "a_clue",
14    "a_correct",
15    "a_definite",
16    "a_detection",
17    "a_different",
18    "a_false",
19    "a_good",
20    "a_guaranteed",
21    "a_hundred",
22    "a_lot",
23    "a_negative",
24    "a_ninety",
25    "a_perfect",
26    "a_positive",
27    "a_rare",
28    "a_result",
29    "a_screening",
30    "a_test",
31    "a_waste",
32    "about",
33    "about_who",
34    "absolute",
35    "absolute_certainty",
36    "absolute_guarantee",
37    "accuracy",
38    "accuracy_is",
39    "accurate",
40    "accurate_test",
41    "actual",
42    "actual_cases",
43    "actual_disease",
44    "after",
45    "after_a",
46    "after_testing",
47    "alarms",
48    "alarms_happen",
49    "alarms_means",
50    "all",
51    "all_groups",
52    "all_positive",
53    "all_positives",
54    "all_screening",
55    "altogether",
56    "altogether_because",
57    "always",
58    "always_harmful",
59    "always_have",
60    "always_mean",
61    "among",
62    "among_everyone",
63    "among_people",
64    "among_positive",
65    "among_positives",
66    "an_absolute",
67    "an_accurate",
68    "an_imperfect",
69    "and_common",
70    "and_false",
71    "and_focus",
72    "and_has",
73    "and_sensitivity",
74    "and_should",
75    "and_specificity",
76    "and_the",
77    "and_there",
78    "answers",
79    "answers_how",
80    "any",
```

```

81  "any_cases",
82  "any_false",
83  "any_mistake",
84  "any_mistakes",
85  "anybody",
86  "are_actual",
87  "are_bad",
88  "are_completely",
89  "are_correct",
90  "are_false",
91  "are_few",
92  "are_identical",
93  "are_missed",
94  "are_no",
95  "are_possible",
96  "are_really",
97  "are_the",
98  "are_true",
99  "as_the",
100 "at_all",
101 "away",
102 "bad",
103 "bad_because",
104 "base",
105 "base_rate",
106 "be_any",
107 "be_false",
108 "be_free",
109 "be_ill",
110 "be_interpreted",
111 "be_left",
112 "be_missed",
113 "be_much",
114 "be_perfect",
115 "be_read",
116 "be_screened",
117 "be_thrown",
118 "be_used",
119 "be_useful",
120 "be_wrong",
121 "because",
122 "because_cases",
123 "because_false",
124 "because_it",
125 "because_the",
126 "because_their",
127 "being_ill",
128 "benefit",
129 "benefit_to",
130 "both",
131 "both_false",
132 "by_all",
133 "calculation",
134 "can_be",
135 "can_give",
136 "can_have",
137 "can_ignore",
138 "can_miss",
139 "can_outnumber",
140 "can_use",
141 "cannot",
142 "cannot_be",
143 "cannot_change",
144 "cannot_ever",
145 "cannot_have",
146 "cannot_help",
147 "cannot_make",
148 "case",
149 "case_should",
150 "cases",
151 "cases_are",
152 "cases_can",
153 "cases_means",
154 "cases_missed",
155 "cases_so",
156 "cases_with",
157 "certain",
158 "certain_proof",
159 "certainty",
160 "certainty_about",
161 "certainty_some",
162 "chance",
163 "chance_after",
164 "chance_among",

```

165 "chance_of",
166 "change",
167 "change_the",
168 "changes",
169 "changes_the",
170 "changing",
171 "changing_how",
172 "changing_the",
173 "clue",
174 "clue_rather",
175 "common",
176 "common_conditions",
177 "common_the",
178 "compare",
179 "compare_detected",
180 "completely",
181 "completely_rule",
182 "completely_useless",
183 "conclusive",
184 "conclusive_and",
185 "condition",
186 "condition_after",
187 "condition_among",
188 "condition_because",
189 "condition_can",
190 "condition_has",
191 "condition_if",
192 "condition_is",
193 "condition_makes",
194 "condition_means",
195 "condition_regardless",
196 "conditions",
197 "conditions_produce",
198 "context",
199 "correct",
200 "correct_positive",
201 "could_be",
202 "count",
203 "count_true",
204 "definite",
205 "definite_diagnosis",
206 "definitely",
207 "definitely_has",
208 "definitely_healthy",
209 "definition",
210 "definition_of",
211 "depends",
212 "depends_on",
213 "detected",
214 "detected_cases",
215 "detection",
216 "detection_rate",
217 "detects",
218 "detects_ninety",
219 "diagnosis",
220 "diagnosis_with",
221 "difference",
222 "different",
223 "different_group",
224 "different_populations",
225 "different_predictive",
226 "directly",
227 "directly_answers",
228 "disease",
229 "disease_chance",
230 "disease_completely",
231 "disease_if",
232 "disease_in",
233 "disease_is",
234 "dismiss",
235 "dismiss_screening",
236 "divided",
237 "divided_by",
238 "do",
239 "do_not",
240 "do_with",
241 "does",
242 "does_not",
243 "effect",
244 "effect_on",
245 "equal",
246 "equal_to",
247 "errors",
248 "errors_show",

```

249 "even",
250 "even_a",
251 "even_one",
252 "ever",
253 "ever_be",
254 "every",
255 "every_population",
256 "every_positive",
257 "every_screening",
258 "everyone",
259 "everyone_who",
260 "everyone_with",
261 "exist",
262 "false",
263 "false_alarms",
264 "false_all",
265 "false_negative",
266 "false_negatives",
267 "false_positive",
268 "false_positives",
269 "false_results",
270 "false_when",
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README.md

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1 # BaseRate
2
3 A field guide to screening uncertainty for UnivaBio 2026. Count a fictional population, compare
  assumptions, predict the answer, then explain your reasoning to a small on-device AI coach.
4
5 [Open the learning lab](https://nicholas-afk.github.io/baserate/) [Demo and project
  materials](https://nicholas-afk.github.io/baserate/project.html)
6
7 ## Run locally
8
9 Serve `dist/` with any static HTTP server: `python3 -m http.server 8000 --directory dist`, then open
  http://localhost:8000. No package installation, API key, account or remote inference is needed. The
  application also works from a local folder; hosting and clipboard features are best used over HTTP.
  Optional downloadable submission materials are included on the hosted project page.
10
11 ## Working features
12
13 - Three fictional test presets, live sliders and editable numeric rates.
14 - A 1,000-person square grid, positive-only focus and natural-frequency tree using the same underlying
  counts.
15 - Exact expected outcomes, explicit denominator and positive predictive value (PPV).
16 - Pin experiment A while varying B; compare outcomes and PPV changes in percentage points.
17 - A prevalence curve using the current test, with an equivalent accessible table.
18 - A conditional second-test explorer: independent identical tests, perfectly repeated results or
  explicit conditional rates. Assumptions are visible; a repeated result can add no information.
19 - Three predict/reveal/denominator cases, first-guess summary and numeric notebook export. No coach
  explanation enters the export.
20 - English teach-back feedback from a trained local text classifier, including influential features,
  coverage, label alternatives, uncalibrated votes and next-practice links.
21 - Conservative uncertainty handling, personal-health query guard and readable missing-model fallback.
22 - Keyboard controls, labelled inputs, patterned outcomes, responsive layout, printable worksheet and
  validated share links containing only fictional rates.
23 - Optional feature-detected WebMCP controls that operate on the same visible state.
24
25 ## Reproduce and verify
26
27 Training uses only Python's standard library. `python3 model/train.py` reproduces the distributed
  model and evaluation. Tests use only Node's standard library:
28
29 ```sh
30 node tests/verify.cjs
31 node tests/extensions.cjs
32 node tests/learning.cjs
33 node tests/display.cjs
34 ```
35
36 The checks cover 336 base arithmetic/rounding combinations, 2,525 prevalence-curve points, 405
  conditional second-test combinations, all three lesson equations, invalid inputs, immutable experiment
  snapshots, undefined denominators, count conservation, precise tiny-count display, text-free exports
  and missing-model feedback. Python/browser inference parity is checked on all 30 holdout examples.
  Manual browser checks covered lesson progression/restart, rate/link validation, comparison pinning,
  conditional presets, local feedback and a 390-pixel layout without horizontal overflow.
37
38 ## AI, data and limitations
39
40 A Multinomial Naive Bayes model uses learned word and adjacent-word features, additive smoothing
  (alpha 0.7), and five educational reasoning labels. The original synthetic dataset contains 100
  English training explanations and 30 separately authored holdout examples; no patient data are used.
41
42 The raw synthetic holdout result is 29/30 correct. Uncertainty checks accept 28/30 explanations,
  with 27/28 correct accepted labels. One correct distinction between sensitivity and PPV is misread.
  See `model/evaluation.json` for complete predictions and confusion matrix. These are small synthetic
  development checks, not an external benchmark, clinical validation, calibrated confidence or evidence
  of learning or health benefit. No holdout sentences were moved into training to hide the known error.
43
44 The model's votes are uncalibrated and may fail on negation, mixed reasoning, unfamiliar language or
  wording. The coach is a suggestion, not an assessment of competence. Its fixed responses are
  inspectable in `dist/coach.js`. Screening arithmetic runs separately from the model. Conditional
  second-test rates apply only among first-positive people; reusing unconditional rates requires the
  displayed independence assumption.
45
46 Only squares are rounded with largest remainders, preserving 1,000 squares. Mathematical expectations
  and PPV use unrounded values. Display precision adapts for small nonzero counts so a defined
  denominator never appears as 0/0.
47
48 ## Privacy
49
50 No analytics, external scripts/fonts, inference requests or stored learner explanations. Text and
  lesson guesses stay in tab memory and disappear on reload. A notebook export contains only numeric
  predictions, case settings and selected denominator choices. Settings links contain only three
```

validated fictional rates. Hosting providers may keep ordinary request logs. Reference links lead to external source sites.

51
52 All rates and cases are fictional. The app does not diagnose, estimate a person's risk, interpret
personal results, select a real test or recommend treatment. The prototype is English only. No
independent learner study, clinical validation or measured health impact has been completed.
53
54 ## Research and design
55
56 Natural frequencies and explicit reference groups informed the grid/tree and denominator-first
teaching. Hierarchy, restrained typography, spacing, meaningful labels and progressive disclosure
informed the editorial interface. Sources support the underlying content and design choices, not this
prototype's effectiveness:
57
58 - [Hoffrage et al., 2015, original natural-frequency
study](https://pmc.ncbi.nlm.nih.gov/articles/PMC4604268/)
59 - [Gigerenzer, BMJ 2011, What are natural frequencies?](https://www.bmj.com/content/343/bmj.d6386)
60 - [National Cancer Institute, Cancer Screening
Overview](https://www.cancer.gov/about-cancer/screening/patient-screening-overview-pdq)
61 - [Nielsen Norman Group, visual design
principles](https://www.nngroup.com/articles/principles-visual-design/)
62 - [GOV.UK Design System, Details component](https://design-system.service.gov.uk/components/details/)
63
64 ## Authorship and next validation
65
66 Code, synthetic data, educational copy, interface and submission materials were created with extensive
AI assistance at the entrant's direction. The demo uses the entrant's supplied recording over
captured states of the actual app, followed by a visual feature tour. No medical expertise, personal
biography, clinical research or learner-testing results are claimed. The entrant must understand the
implementation and be ready to explain it to judges.
67
68 Next steps: independently collected learner explanations, health-educator review, evaluated
multilingual support, accessibility testing with users and a properly designed comprehension study.
These are proposed work, not completed outcomes.
69
70 ## License
71
72 MIT for original application source and synthetic dataset. Linked research retains its own rights. The
entrant-provided narration is used for this competition demo; the source license does not grant
rights to impersonate the entrant or reuse their voice.

SUBMISSION.md

```
1  ## Inspiration
2
3  A test that detects 90% of cases can still produce a positive group in which most results are false
  positives. The denominator explains the apparent contradiction. BaseRate turns that distinction into a
  learning experiment: predict, count, change an assumption and explain it back.
4
5  The audience is students, health-literacy learners and educators. The intended contribution to human
  health is understanding screening uncertainty. The project does not diagnose or estimate personal
  risk. [Natural-frequency research](https://pmc.ncbi.nlm.nih.gov/articles/PMC4604268/) informed its
  representations, and the [National Cancer
  Institute](https://www.cancer.gov/about-cancer/screening/patient-screening-overview-pdq) informs its
  screening context. These sources do not validate this prototype.
6
7  ## What it does
8
9  The working lab connects several views of the same fictional population. Change prevalence,
  sensitivity and specificity with sliders or numeric inputs. A 1,000-person grid, positive-group focus,
  frequency tree, four expected outcome counts and PPV equation update together. Only the squares are
  rounded.
10
11 Pin experiment A and vary B to compare the positive groups. A prevalence curve, with an equivalent
  table, shows why the same test changes meaning across populations. In the default case, 1% prevalence,
  90% sensitivity and 95% specificity give 9 true positives and 49.5 false positives:  $9/58.5 = 15.4\%$ .
  At 20% prevalence, the unchanged test gives 81.8%.
12
13 The second-test explorer applies explicit conditional rates only to first-positive people. Independent
  identical tests yield 76.6% in the default fictional example. A perfectly repeated result leaves PPV
  unchanged. The assumptions are visible so the tool does not imply that two positives automatically
  provide independent evidence.
14
15 A three-case lesson asks for an estimate before revealing counts, then asks which group belongs in the
  denominator. First guesses stay in tab memory; learners can export a numeric notebook. A
  demonstration estimate is explicitly labelled as an example, not learner-study data.
16
17 The teach-back AI coach reads an explanation locally, identifies a likely reasoning pattern and
  selects fixed educational feedback, current experiment counts, a follow-up question and evidence
  links. Learners can inspect influential words, feature coverage, label alternatives and uncalibrated
  model votes, then follow a relevant practice link. The coach suggests an interpretation rather than
  grading competence.
18
19 ## How it was built
20
21 A static HTML/CSS/JavaScript app runs without dependencies, accounts or API keys. Pure arithmetic and
  learning modules feed separate interface controllers. A dependency-free Python trainer learns a
  Multinomial Naive Bayes classifier from 100 synthetic English explanations. Word and adjacent-word
  features distinguish contextual reasoning and four common misconception patterns.
22
23 Inference stays in the browser. Coverage and vote separation can trigger uncertainty; personal-health
  queries are redirected away from the fictional experiment. No analytics, remote inference or stored
  explanations are included. Hosting providers may retain ordinary request logs.
24
25 The interface uses a restrained field-notebook style, clear hierarchy, readable labels, patterned
  outcomes and keyboard controls. Essential explanations stay visible; advanced assumptions and model
  details use progressive disclosure. The demo uses the entrant's own supplied recording and actual
  captured application states.
26
27 ## Challenges and what was learned
28
29 Exact expectations can be fractional while the grid must contain exactly 1,000 squares.
  Largest-remainder rounding keeps that illustration consistent; calculations, comparisons and practice
  answers retain raw values. Adaptive display precision preserves tiny nonzero denominators.
30
31 Repeated testing exposed a second conceptual trap: multiplying test rates without explaining
  dependence. The conditional explorer makes the selected population and assumption explicit, including
  a repeat that adds no information.
32
33 A small text model can sound overly certain or misread a correct explanation. The app exposes
  uncertainty and failure modes. One holdout sentence correctly distinguishing sensitivity from
  predictive value was misread as confusion. The project retains the known error and complete evaluation
  instead of moving the sentence into training.
34
35 ## Validation
36
37 Four reproducible test suites pass. They cover 336 base arithmetic/rounding combinations, 2,525
  prevalence-curve points, 405 conditional repeat-test combinations, all lesson equations, immutable
  snapshots, invalid inputs, zero denominators, tiny-count display, exports excluding learner text and
  missing-model feedback. All 30 holdout examples match between Python and browser inference.
38
39 Browser checks verified comparison pinning, independent/repeated presets, three-case progression and
  restart, numeric-input errors and recovery, validated settings links, local model diagnostics and the
```

actual app at 390 pixels without horizontal overflow. A separate code review was completed and its display/recovery findings corrected.

40

41 The separately authored synthetic holdout produced 29/30 correct raw classifications. Uncertainty checks accepted 28 labels, 27 correct. These are development checks, not a benchmark, clinical accuracy claim, evidence of learning gain or health outcome. No independent learner study or clinical validation has been completed.

42

43 ## AI assistance and honest limits

44

45 The project, interface, synthetic dataset, implementation and submission materials were created with extensive AI assistance at the entrant's direction. The entrant supplied the narration and must understand and explain the code. No clinical research, medical expertise or user testing is claimed.

46

47 All scenarios are fictional. The English-only classifier can fail on negation, mixed ideas, unfamiliar phrasing and other languages. Model votes are uncalibrated. The app does not interpret personal results, diagnose, recommend real tests or give treatment advice. The AI neither computes the screening probability nor generates medical facts.

48

49 ## What's next

50

51 Independent learner explanations, health-educator review, accessibility testing with users, evaluated multilingual support and a properly designed comprehension study. These are proposed validation steps, not completed outcomes.

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```
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2
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